D09-R Revised Visual Validation Report: GPIM Layout Repair and NRC Service-Life Interpretation

Capacity Utilization US-Chile Source-of-Truth Pipeline

2026-07-01

Contents

1 Purpose and How to Read This Report

This D09-R report repairs the human-facing D09 validation report. It exists because the original D09 artifacts passed validation but left too much interpretive work to the reader: figures drifted away from their sections, captions were thin, and the NRC real-stock path needed a direct boundary interpretation.

Read the report as a visual validation interface. The figures reproduce or reuse D09 figure data and D09 figures; they do not create a new source-of-truth panel. The central repair is the placement of each figure next to the paragraph that interprets it.

D09-R does not start D10 transformation planning. It does not rebuild GPIM, change survival parameters, change pKN, run sensitivity analysis, run econometrics, or authorize model transformations.

2 Locked Lineage and Report-Only Boundary

The lineage remains locked through D09. D07 supplies the source-of-truth level/accounting panel, D08 audits that panel and carries the NRC warmup flag, and D09 supplies report-only visual validation outputs.

The revised report preserves the same boundary language: REPORT_ONLY, INSPECTION_ONLY, NOT_SOURCE_OF_TRUTH, NOT_MODEL_READY, and REQUIRES_D10_AUTHORIZATION_FOR_TR
The evidence supports a layout repair, not a data reconstruction.

No total capital, total fixed assets, IPP, residential capital, government transportation, or unvalidated financial correction is promoted as a baseline object.

3 GPIM Equations and Baseline Capital-Stock Architecture

This section states the baseline accounting architecture before the figures appear. The baseline capital object remains D06/D07 ME plus NRC capacity capital, with real stocks, current-cost stocks, and pKN valuation support kept distinct.

The equations clarify why current-cost figures can diverge from real-stock figures: current-cost stocks combine the real gross surviving stock with pKN valuation. That distinction matters for interpreting the NRC path.

The equations are displayed for orientation only. D09-R does not re-estimate, refreeze, or transform any variable.

$$I^R_{j,t} = I^C_{j,t}/(p^K_{j,t}/100) \tag{1}$$

$$K^R_{j,t} = \sum_{a=0}^{A} s_j(a) I^R_{j,t-a} \tag{2}$$

$$K^C_{j,t} = K^R_{j,t} p^K_{j,t}/100 \tag{3}$$

$$K^R_{cap,t} = K^R_{ME,t} + K^R_{NRC,t} \tag{4}$$

$$K^C_{cap,t} = K^C_{ME,t} + K^C_{NRC,t} \tag{5}$$

4 GPIM Real Stock Levels: Accumulation Tendency and NRC Decline

This section places the real-stock level and index figures together because the NRC concern begins in the real gross surviving GPIM object. Figures \{}ref{fig:D09_fig01_real_gpim_stock_levels}, \{}ref{fig:D09_fig03_real_gpim_stock_levels_logscale}, \{}ref{fig:D09_fig05_real_gpim_indices_1947_100},

\{}ref{fig:D09_fig06_real_gpim_indices_2017_100} show the real ME, NRC, and capacity paths before any current-cost valuation effects enter.

The decline or stagnation of NRC real gross GPIM stock is not interpreted as literal physical demolition of all nonresidential structures. The object is nonfinancial-sector productive-capacity nonresidential construction. A structure can remain physically standing while leaving the nonfinancial productive-capacity boundary through abandonment, rezoning, conversion to residential use, or absorption into financial/real-estate uses. This interpretation is plausible but remains subject to service-life and warmup sensitivity review.

This section does not authorize using the NRC real-stock path as a transformed model variable. It only states the boundary-safe interpretation required for human validation.

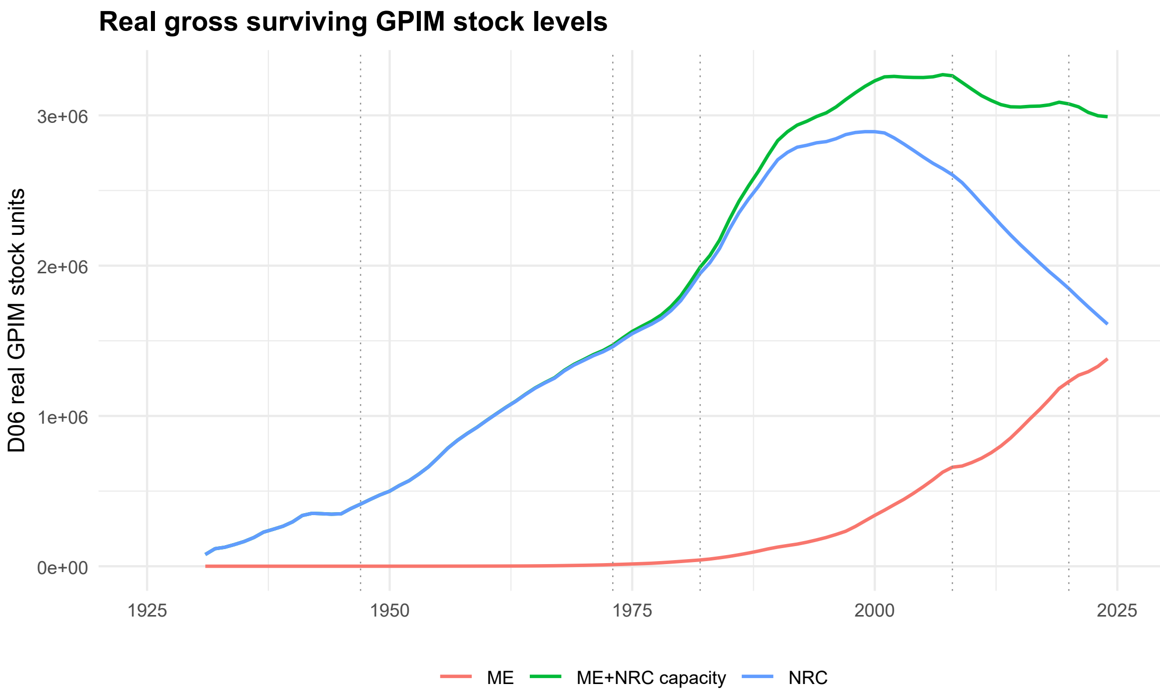

Figure 1: Real gross surviving GPIM stock levels. Baseline D06/D07 refrozen real gross surviving GPIM stocks or report-only visual indices from D09 figure data. The object is source-of-truth only at the underlying level variables; the index view is REPORT_ONLY and INSPECTION_ONLY. The NRC path is interpreted as a nonfinancial productive-capacity boundary object, not a literal census of every standing building.

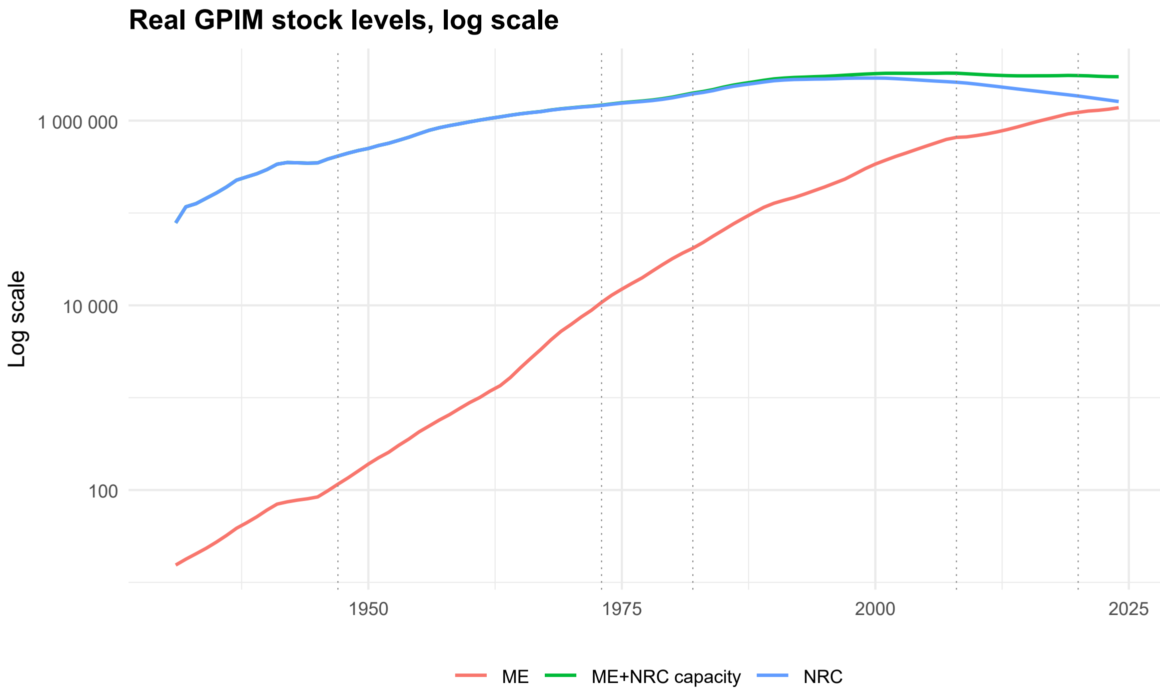

Figure 2: Real gross surviving GPIM stock levels, log scale. Baseline D06/D07 refrozen real gross surviving GPIM stocks or report-only visual indices from D09 figure data. The object is source-of-truth only at the underlying level variables; the index view is REPORT_ONLY and INSPECTION_ONLY. The NRC path is interpreted as a nonfinancial productive-capacity boundary object, not a literal census of every standing building.

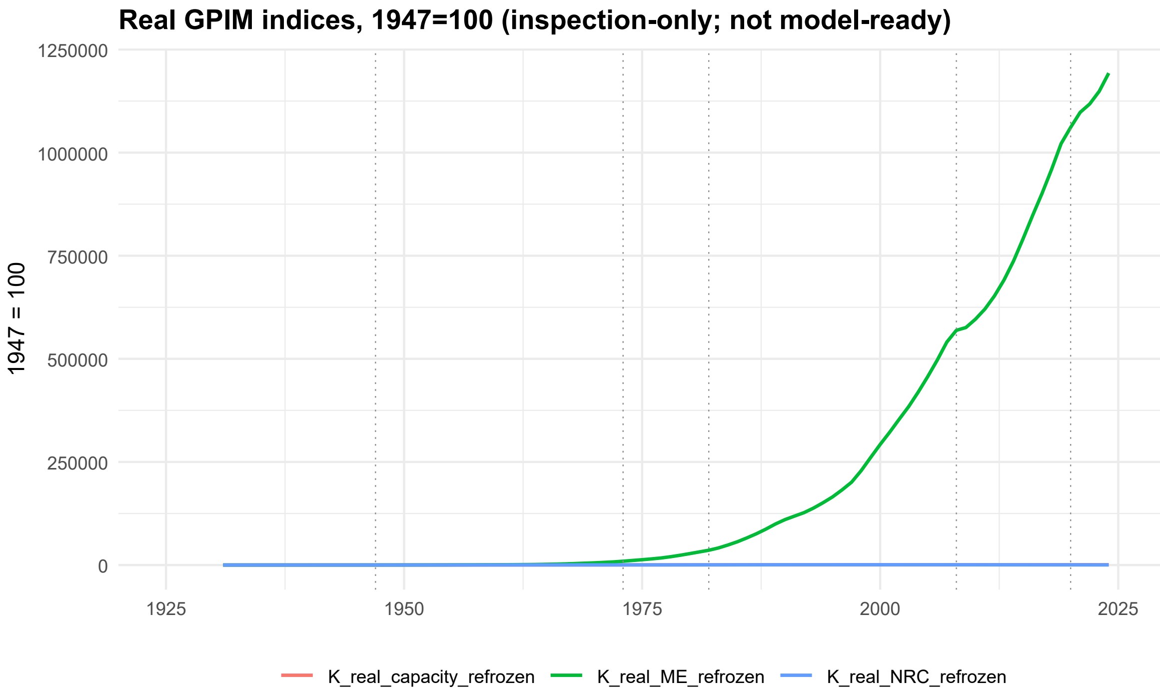

Figure 3: Real GPIM visual indices, 1947=100. Baseline D06/D07 refrozen real gross surviving GPIM stocks or report-only visual indices from D09 figure data. The object is source-of-truth only at the underlying level variables; the index view is REPORT_ONLY and INSPECTION_ONLY. The NRC path is interpreted as a nonfinancial productive-capacity boundary object, not a literal census of every standing building.

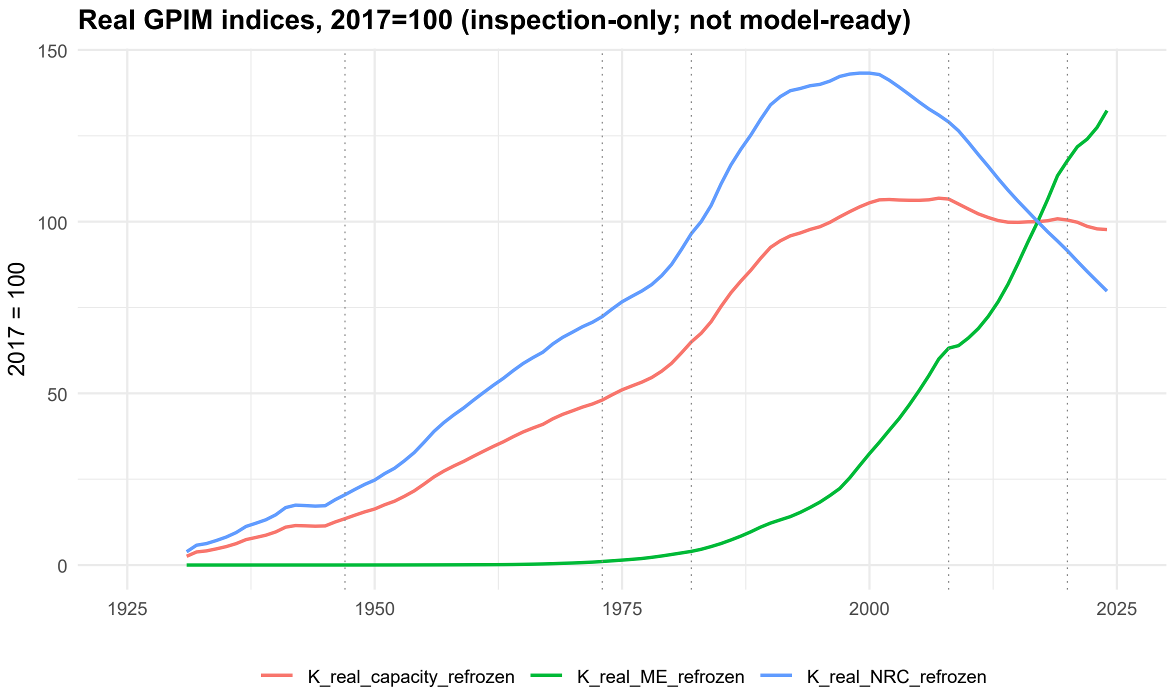

Figure 4: Real GPIM visual indices, 2017=100. Baseline D06/D07 refrozen real gross surviving GPIM stocks or report-only visual indices from D09 figure data. The object is source-of-truth only at the underlying level variables; the index view is REPORT_ONLY and INSPECTION_ONLY. The NRC path is interpreted as a nonfinancial productive-capacity boundary object, not a literal census of every standing building.

5 GPIM Current-Cost Stock Levels and Valuation Interpretation

This section separates current-cost stocks from real stocks. Figures \{}ref{fig:D09_fig02_current_cost_gpim_stock_
\{}ref{fig:D09_fig04_current_cost_gpim_stock_levels_logscale}, \{}ref{fig:D09_fig07_current_cost_gpim_indi
\{}ref{fig:D09_fig08_current_cost_gpim_indices_2017_100} show levels and report-only indices after pKN valuation enters the accounting object.

Current-cost paths mix quantity and valuation effects. A current-cost increase can coexist with real-stock stagnation or decline if pKN movements offset real quantity movements; the interpretation therefore cannot be read as a pure physical-capacity path.

This section does not change pKN and does not use current-cost paths as a substitute for real-capacity validation.

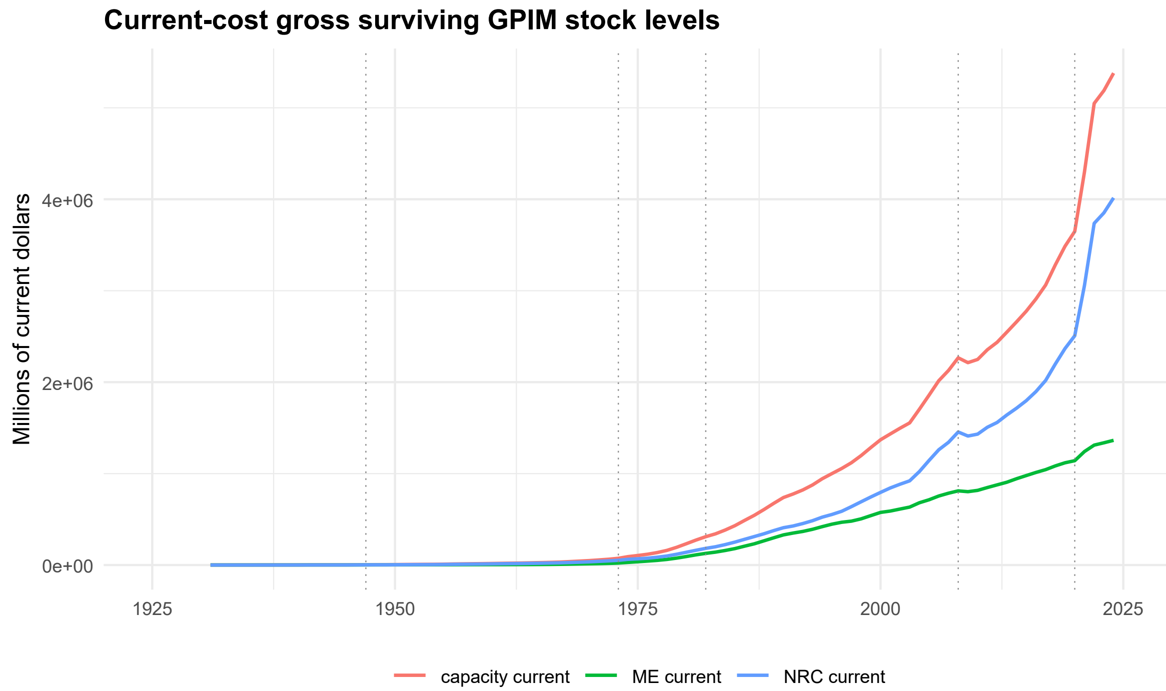

Figure 5: Current-cost gross surviving GPIM stock levels. Baseline D06/D07 current-cost GPIM levels or D09 report-only current-cost visual indices. These paths mix quantity and valuation effects through pKN support and are not model transformations.

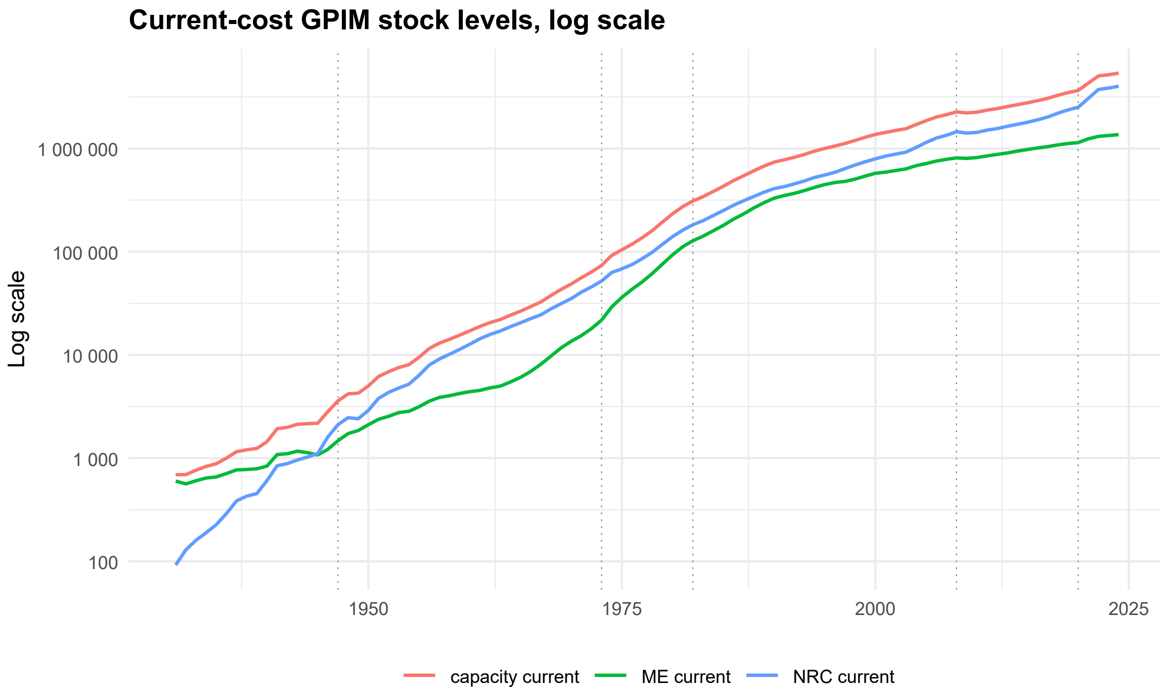

Figure 6: Current-cost gross surviving GPIM stock levels, log scale. Baseline D06/D07 current-cost GPIM levels or D09 report-only current-cost visual indices. These paths mix quantity and valuation effects through pKN support and are not model transformations.

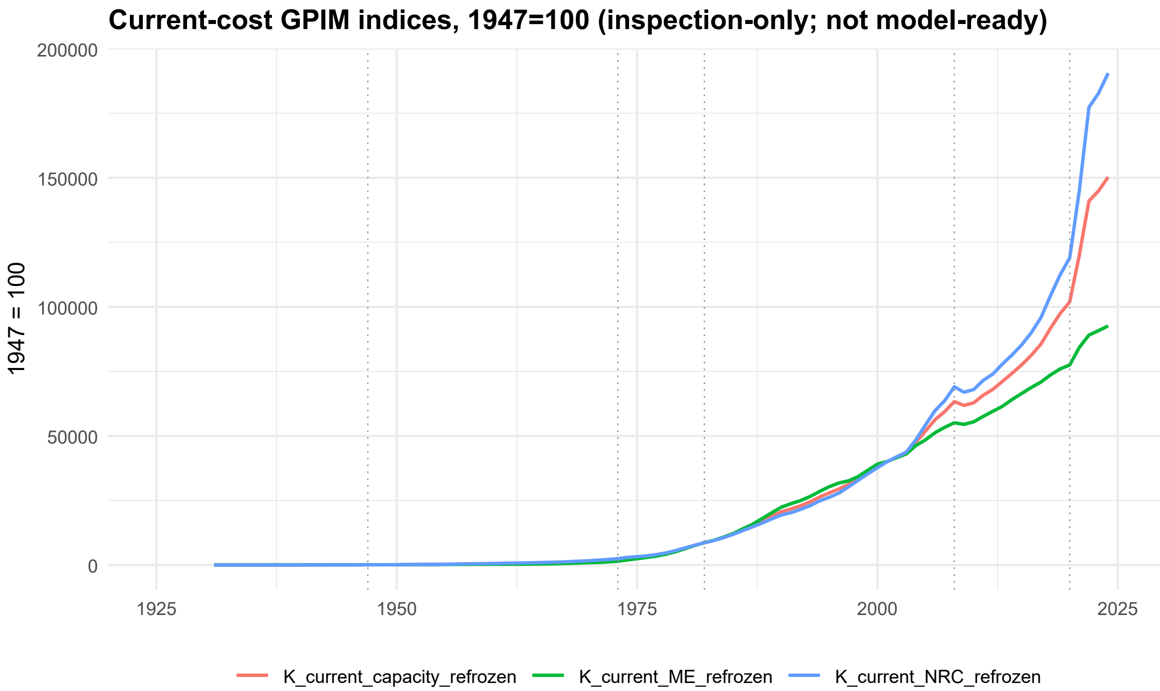

Figure 7: Current-cost GPIM visual indices, 1947=100. Baseline D06/D07 current-cost GPIM levels or D09 report-only current-cost visual indices. These paths mix quantity and valuation effects through pKN support and are not model transformations.

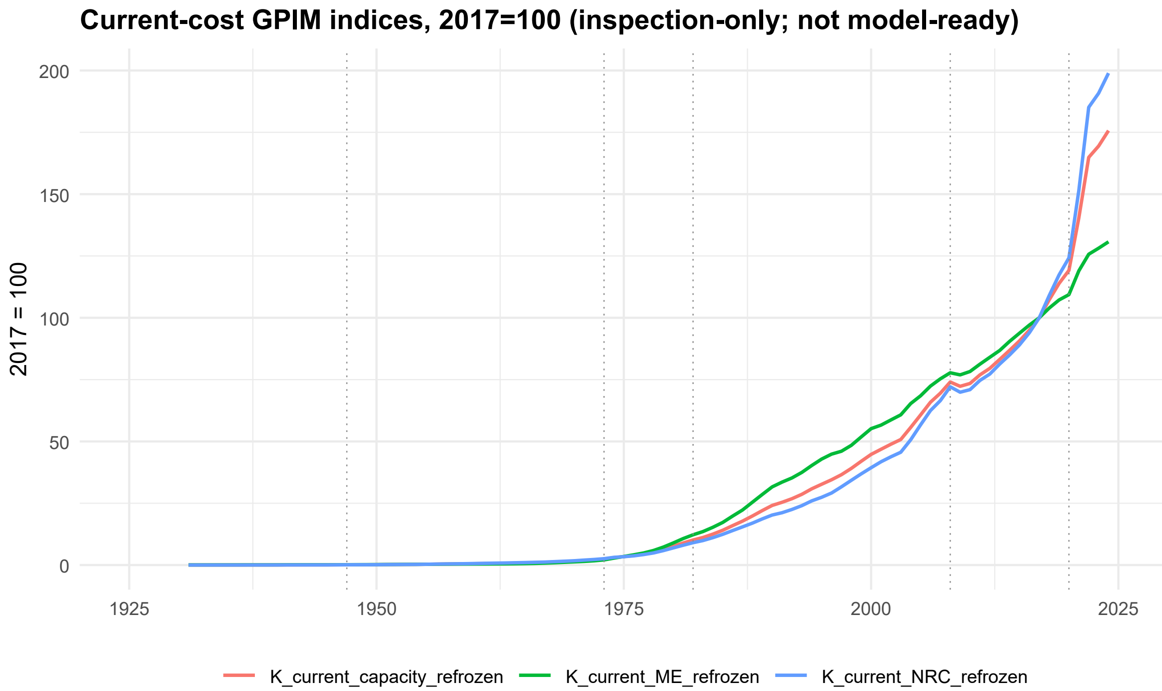

Figure 8: Current-cost GPIM visual indices, 2017=100. Baseline D06/D07 current-cost GPIM levels or D09 report-only current-cost visual indices. These paths mix quantity and valuation effects through pKN support and are not model transformations.

6 Warmup, Survival Profiles, and the NRC Service-Life Concern

This section places warmup and survival directly before the identity checks because the NRC concern is a service-life and initialization concern. Figures \{}ref{fig:D09_fig09_gpim_warmup_timeline}, \{}ref{fig:D09_fig10_locked_survival_profiles_ME_NRC} show the warmup timeline and locked survival profiles.

ME warmup is likely adequate relative to L=14. NRC warmup is review-required relative to L=30. The review flag is non-blocking in D08/D09 because identities, pKN normalization, real-investment scale, current-cost valuation, and source-consumption checks pass.

This section does not change L=30 or alpha=1.6 for NRC. It records the concern and points to future sensitivity design.

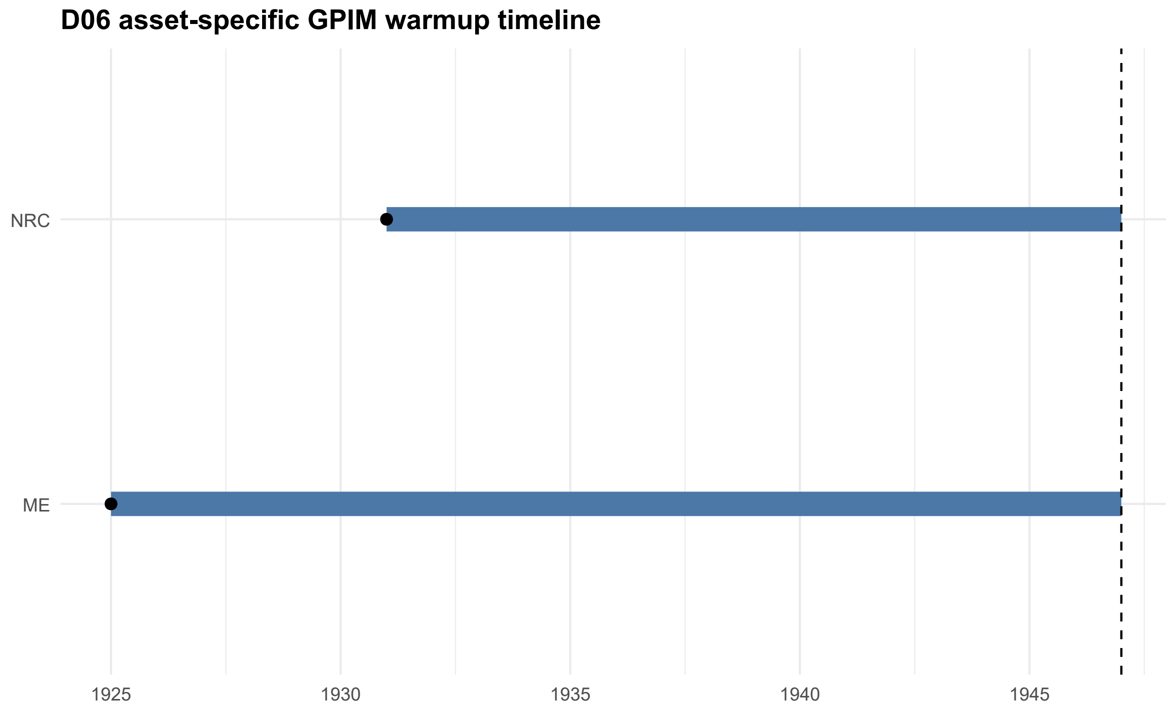

Figure 9: Asset-specific GPIM warmup timeline. D08/D09 diagnostic figure using locked D06 survival parameters. ME L=14 is likely adequate after warmup; NRC L=30 remains REVIEW_REQUIRED. D09-R does not change L, alpha, warmup, or GPIM stocks.

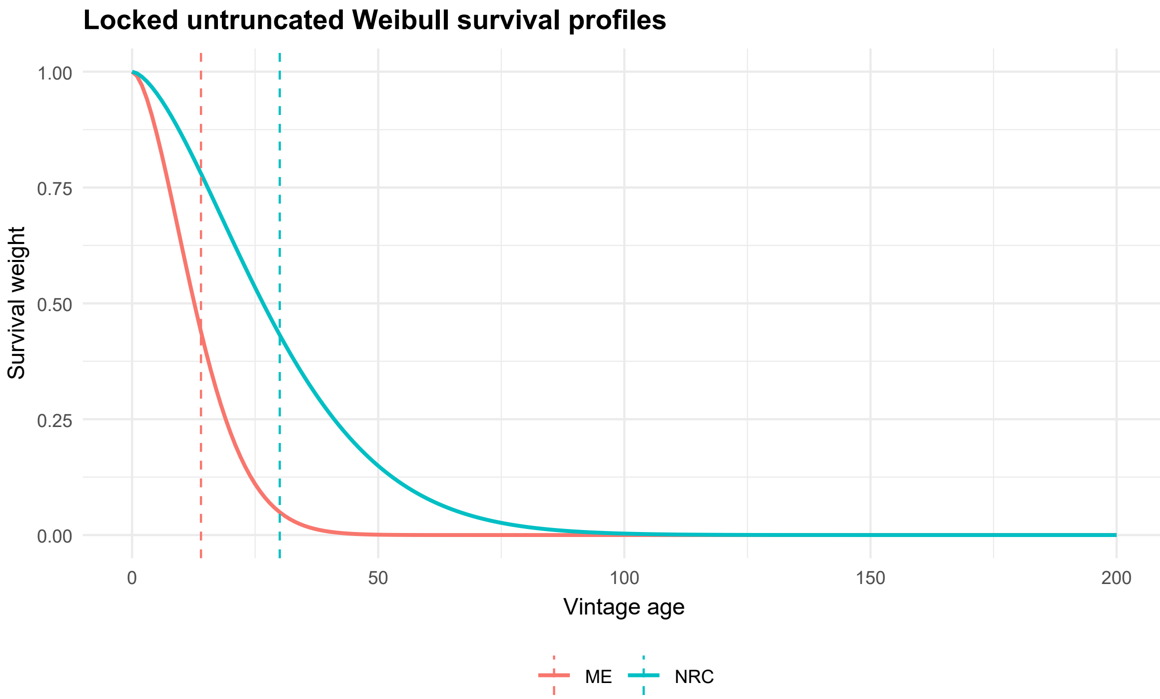

Figure 10: Locked untruncated Weibull survival profiles. D08/D09 diagnostic figure using locked D06 survival parameters. ME L=14 is likely adequate after warmup; NRC L=30 remains REVIEW_REQUIRED. D09-R does not change L, alpha, warmup, or GPIM stocks.

asset	warmup_length	survival_L	survival_alpha	warmup_to_L_ratio	sufficiency_status
ME	22	14	1.7	1.5714286	LIKELY_ADEQUATE
NRC	16	30	1.6	0.5333333	REVIEW_FLAG_SHORT_I

issue_id	risk_level	blocking_stat
NRC_REAL_GROSS_STOCK_DECLINE	MEDIUM	REVIEW_RE
NRC_WARMUP_SHORT_RELATIVE_TO_L30	MEDIUM	REVIEW_RE
NRC_L30_SERVICE_LIFE_UNCERTAINTY	MEDIUM	REVIEW_RE
SECTOR_BOUNDARY_NONFINANCIAL_STRUCTURES	MEDIUM	REVIEW_RE
PHYSICAL_BUILDING_STOCK_VS_PRODUCTIVE_CAPACITY_STOCK	MEDIUM	REVIEW_RE
REZONING_CONVERSION_ABANDONMENT_INTERPRETATION	MEDIUM	REVIEW_RE
FINANCIAL_NONPRODUCTIVE_ABSORPTION_INTERPRETATION	MEDIUM	REVIEW_RE
FUTURE_SENSITIVITY_NEED	MEDIUM	REVIEW_RE

7 Stock-Flow and Current-Cost Identity Audits

This section comes before composition and ratio plots because accounting coherence must be visible before derived diagnostics. Figures \{}ref{fig:D09_fig11_real_investment_scale_residuals}, \{}ref{fig:D09_fig12_cu \{}ref{fig:D09_fig13_capacity_identity_residuals} display the real-investment scale, current-cost valuation, and capacity aggregation residuals.

The residual evidence supports the D08 conclusion that the GPIM repair did not break the required identities. This is why the NRC service-life issue remains review-required rather than blocking at D09-

R.

These plots are audits, not regressions or transformation diagnostics. They do not authorize D10 use.

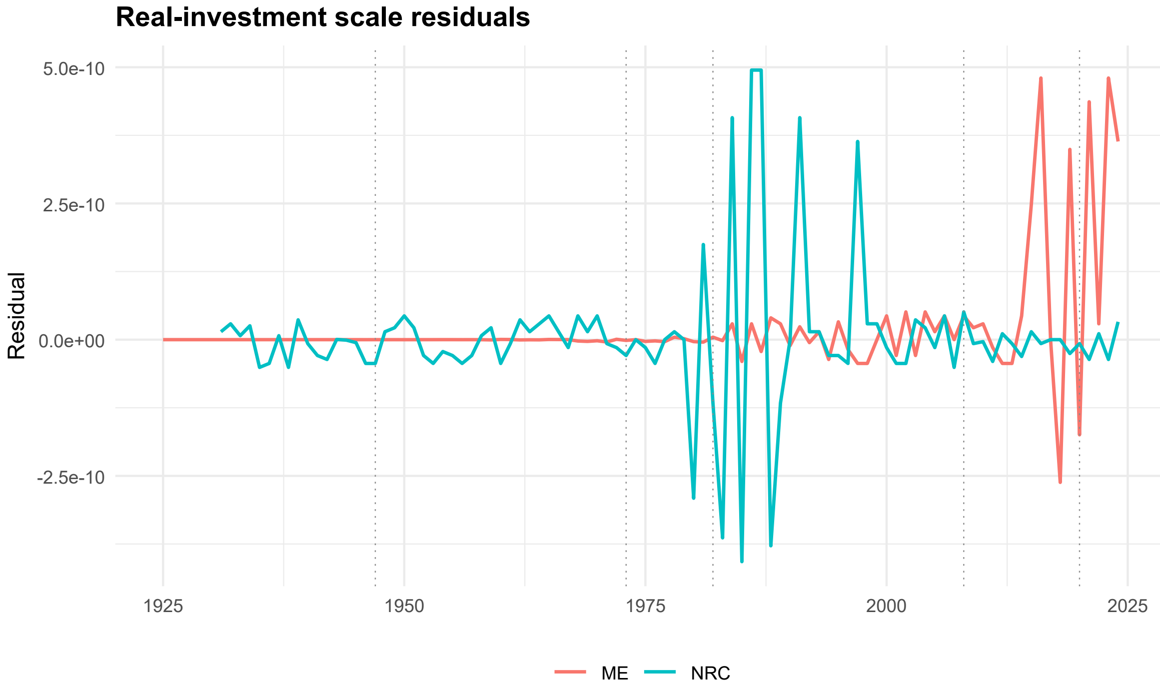

Figure 11: Real-investment scale residuals. D08/D09 audit diagnostic showing identity or scale residuals. The plotted object is an accounting validation diagnostic, not an econometric result and not a new source-of-truth variable.

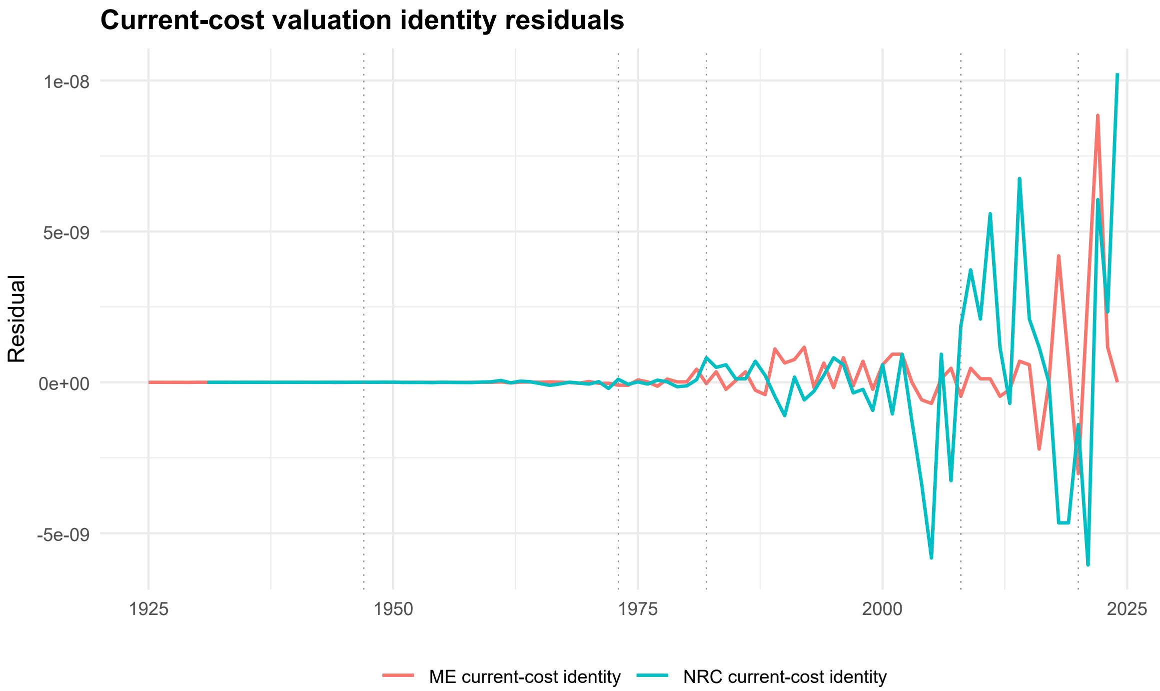

Figure 12: Current-cost valuation identity residuals. D08/D09 audit diagnostic showing identity or scale residuals. The plotted object is an accounting validation diagnostic, not an econometric result and not a new source-of-truth variable.

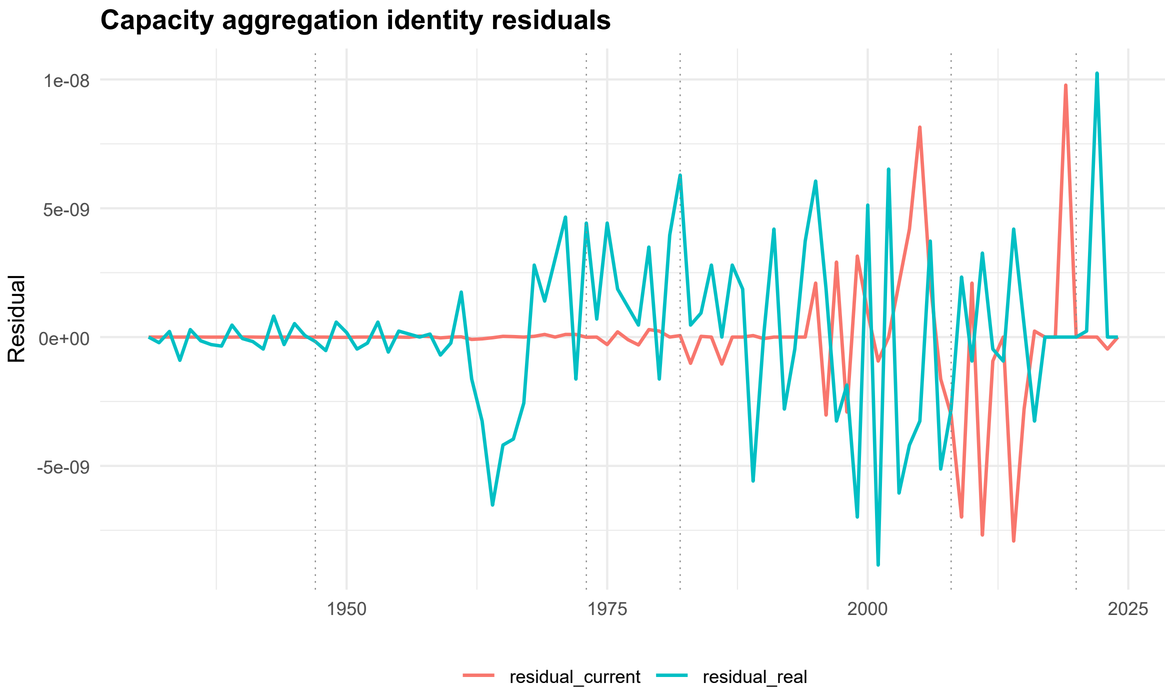

Figure 13: Capacity aggregation identity residuals. D08/D09 audit diagnostic showing identity or scale residuals. The plotted object is an accounting validation diagnostic, not an econometric result and not a new source-of-truth variable.

object	metric	max_abs_residual	audit_s
ME	I_real residual	0	PASS
NRC	I_real residual	0	PASS
K_current_ME_refrozen_identity	current-cost identity	0	PASS
K_current_NRC_refrozen_identity	current-cost identity	0	PASS
K_current_capacity_refrozen_identity	current-cost identity	0	PASS
K_real_capacity_refrozen_equals_ME_plus_NRC	capacity identity	0	PASS
K_current_capacity_refrozen_equals_ME_plus_NRC	capacity identity	0	PASS

8 ME/NRC Composition and Mechanization Diagnostics

This section groups the ME/NRC ratios and shares so the reader can see composition after the level and identity evidence. Figures \{}ref{fig:D09_fig14_ME_over_NRC_real_stock_ratio}, \{}ref{fig:D09_fig15_ME_o \{}ref{fig:D09_fig16_ME_NRC_real_capacity_shares}, \{}ref{fig:D09_fig17_ME_NRC_current_capacity_sha \{}ref{fig:D09_fig18_ME_over_NRC_real_investment_ratio}, \{}ref{fig:D09_fig19_ME_over_NRC_current_ compare real, current-cost, and investment composition.

The composition figures make the mechanization and construction mix visible without creating a new source variable. They are useful precisely because they are downstream inspection objects with the baseline ME/NRC boundary intact.

No composition ratio is model-ready in D09-R. Any use as q or another transformation requires later D10 authorization.

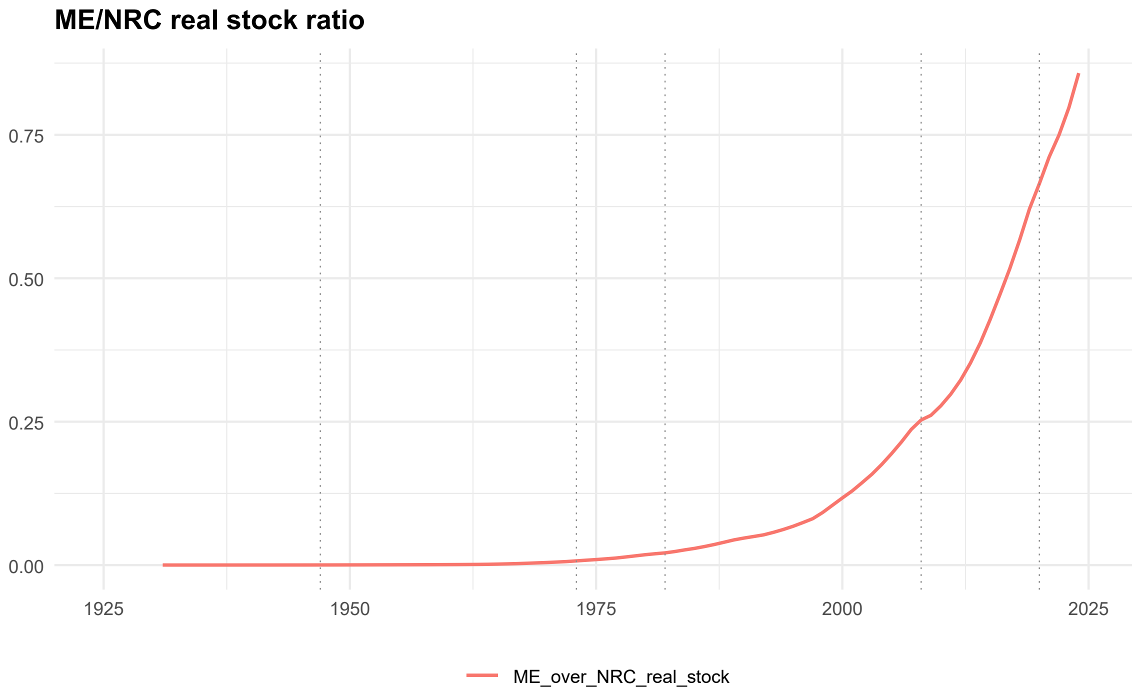

Figure 14: ME/NRC real stock ratio. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

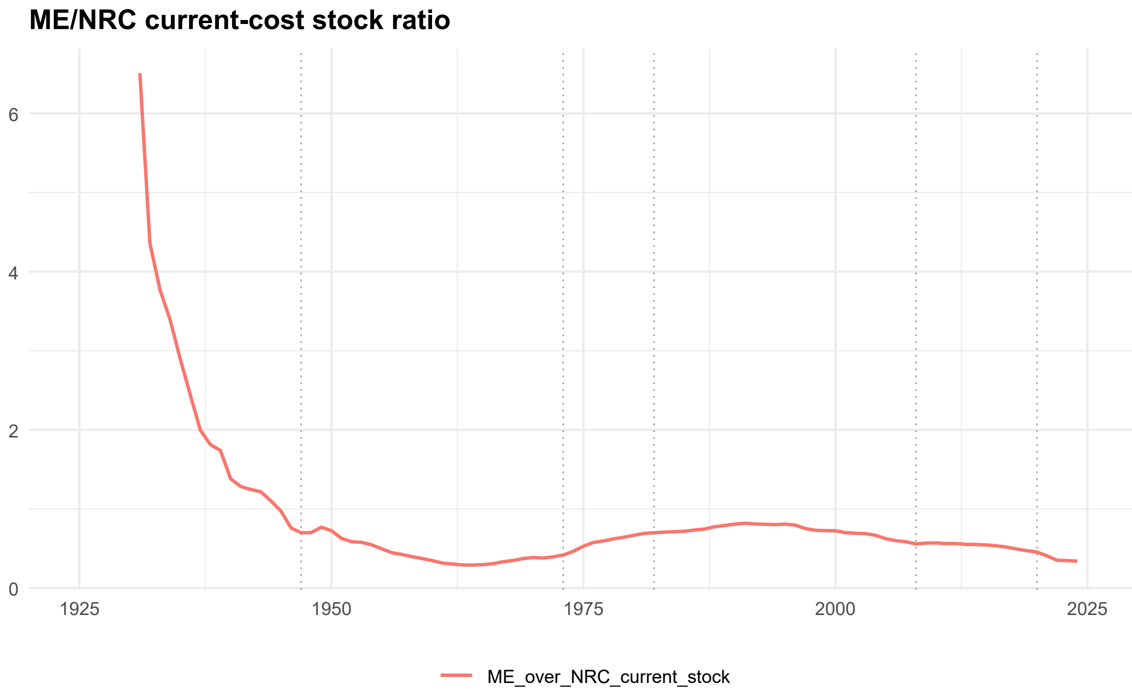

Figure 15: ME/NRC current-cost stock ratio. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

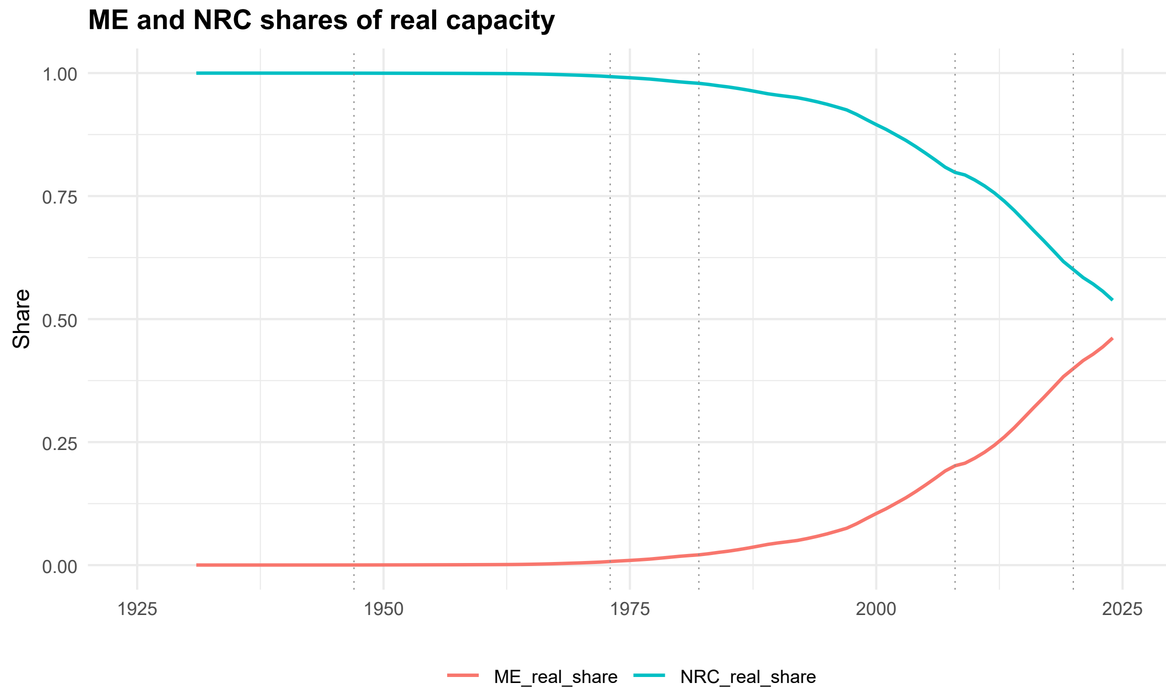

Figure 16: ME and NRC shares of real capacity. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

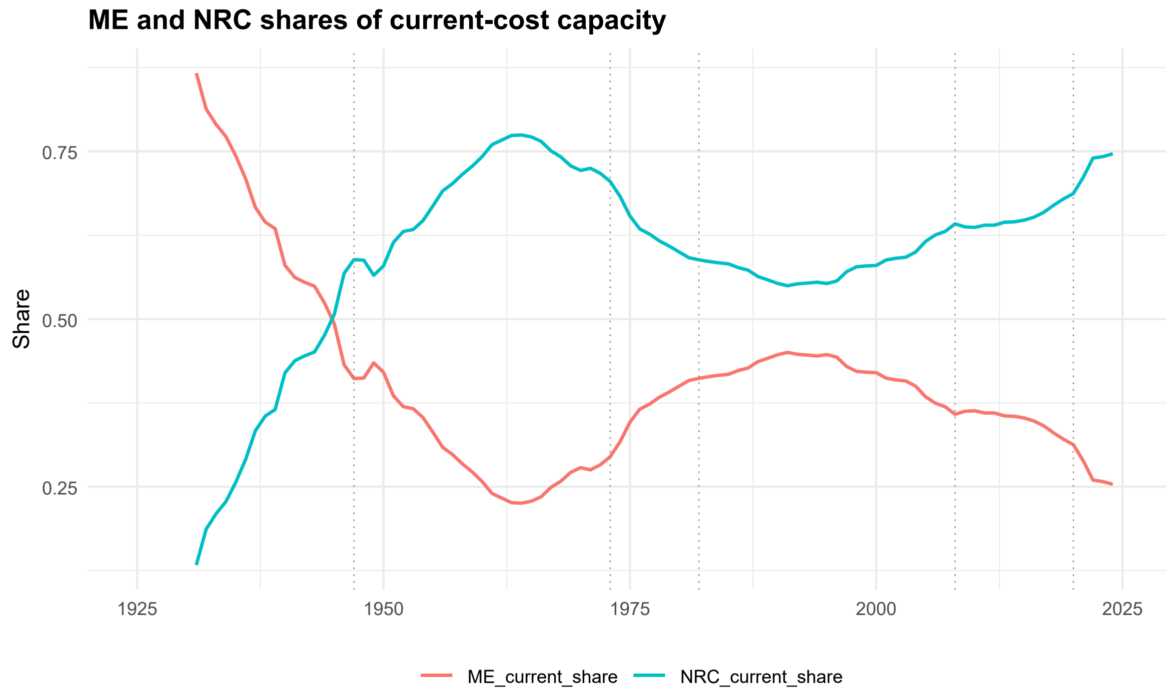

Figure 17: ME and NRC shares of current-cost capacity. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

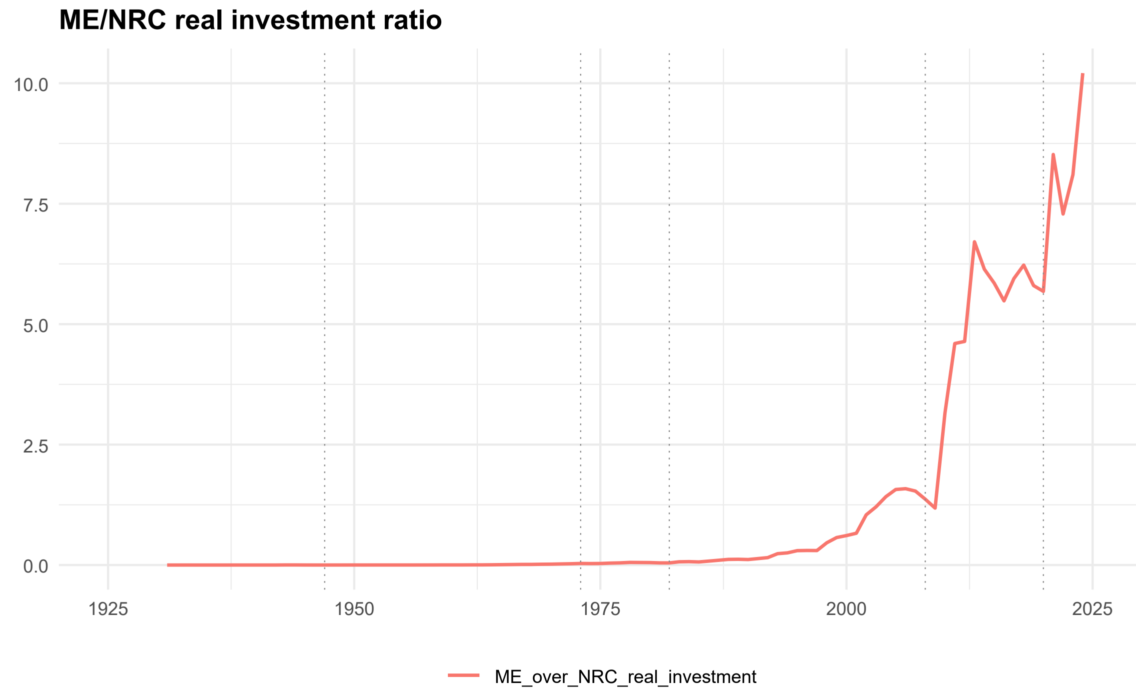

Figure 18: ME/NRC real investment ratio. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

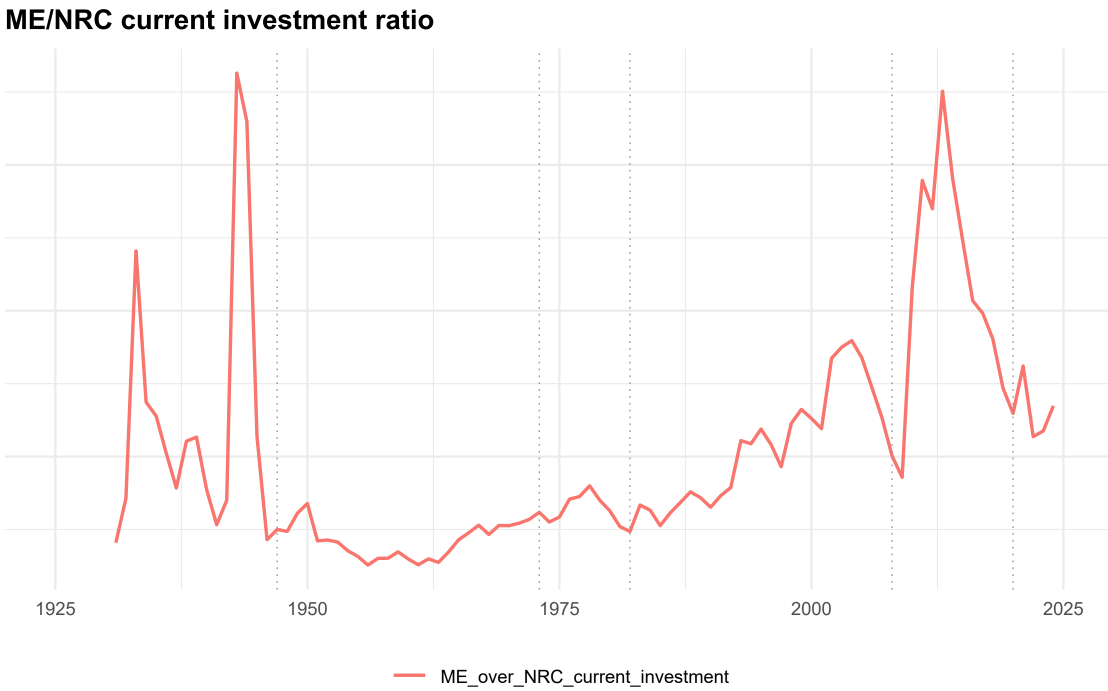

Figure 19: ME/NRC current investment ratio. Report-only ME/NRC composition diagnostic from D09 figure data. It compares D06/D07 baseline ME and NRC components inside the nonfinancial capacity-capital boundary and is not a model transformation unless D10 later authorizes it.

9 Net-over-Gross Comparisons and Concept Boundaries

This section isolates net-over-gross diagnostics from baseline capital measures. Figures \{}ref{fig:D09_fig20_net_ove \{}ref{fig:D09_fig21_net_over_gross_current_cost_NRC}, \{}ref{fig:D09_fig22_net_over_gross_current_cost compare current-cost net-stock concepts with current-cost gross GPIM stocks where those comparisons are available.

Net-over-gross comparisons are accounting diagnostics, not substitutes for the D06 gross surviving GPIM baseline. They compare current-cost net stock concepts to current-cost gross GPIM stocks where available.

The section does not replace the baseline with net stocks and does not promote accounting comparisons to source-of-truth status.

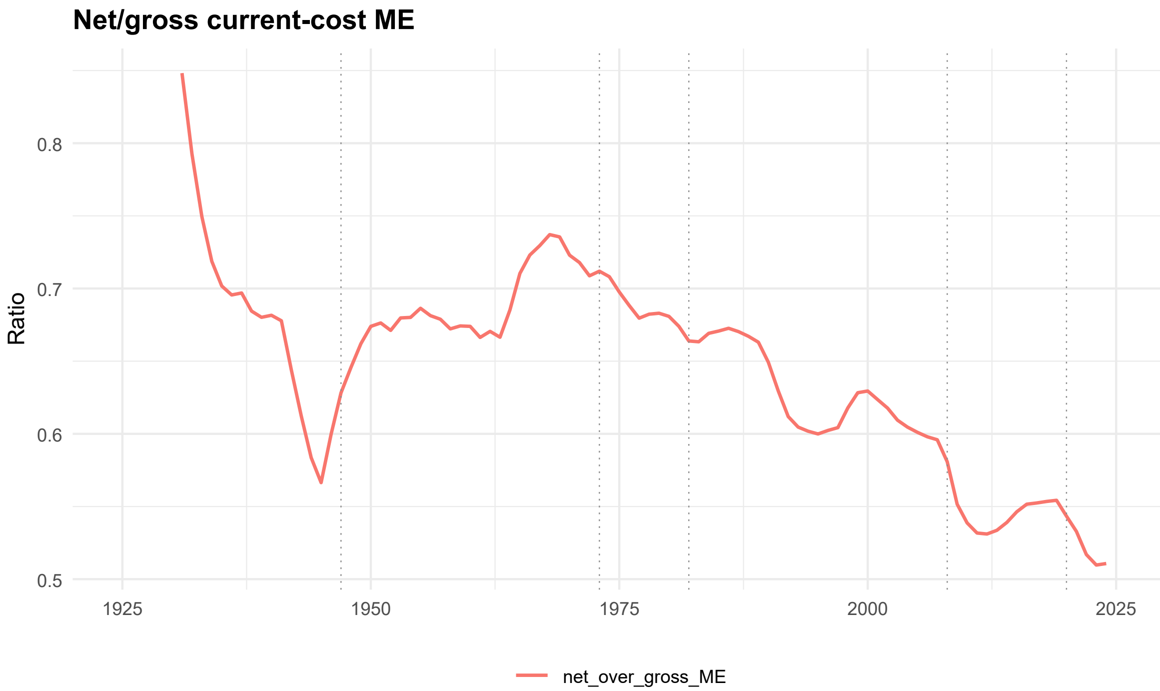

Figure 20: Net/gross current-cost ME. Accounting-comparison diagnostic from D09 figure data. Net-over-gross comparisons are not substitutes for the D06 gross surviving GPIM baseline; they compare current-cost net stock concepts to current-cost gross GPIM stocks where available.

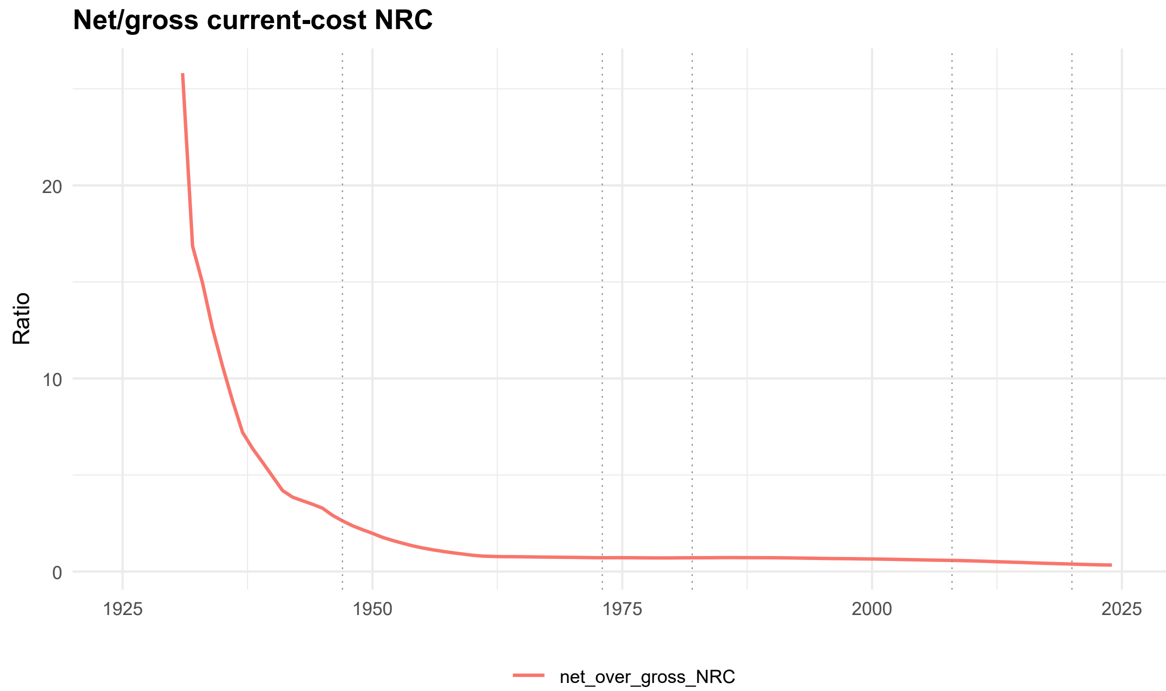

Figure 21: Net/gross current-cost NRC. Accounting-comparison diagnostic from D09 figure data. Net-over-gross comparisons are not substitutes for the D06 gross surviving GPIM baseline; they compare current-cost net stock concepts to current-cost gross GPIM stocks where available.

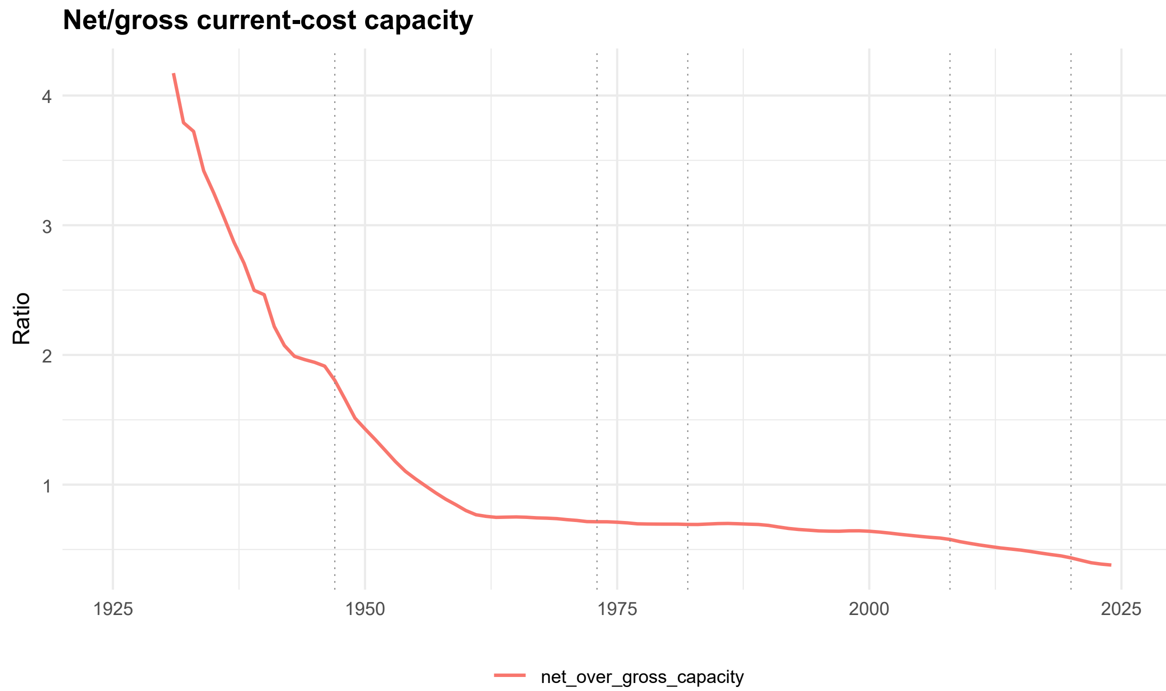

Figure 22: Net/gross current-cost capacity. Accounting-comparison diagnostic from D09 figure data. Net-over-gross comparisons are not substitutes for the D06 gross surviving GPIM baseline; they compare current-cost net stock concepts to current-cost gross GPIM stocks where available.

10 Capital-Measure Status Map: Baseline, Comparison, Review-Only, Parked, Excluded

This section repairs the status-map flow by placing the status map, baseline comparison, and parked/excluded view together. Figures \{}ref{fig:D09_fig23_capital_measure_status_map}, \{}ref{fig:D09_fig24_baseline_vs_a \{}ref{fig:D09_fig25_parked_excluded_capital_objects_appendix} show which capital objects are baseline, comparison, review-only, parked, or excluded.

The parked/excluded figure is kept in the main status section only as a boundary-warning device. It prevents the reader from mistaking broader fixed-asset objects for the baseline ME+NRC capacity object.

Parked and excluded objects are not visually promoted as alternatives to baseline. They remain outside transformation use.

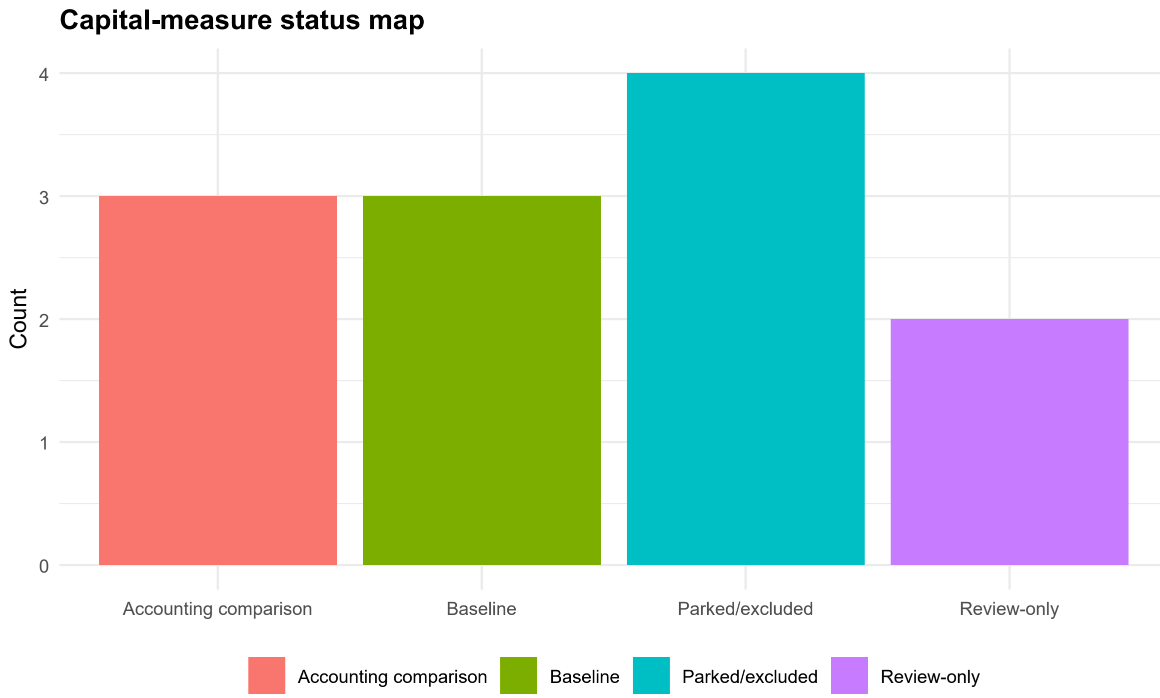

Figure 23: Capital-measure status map. Boundary-status diagnostic separating baseline, accounting-comparison, review-only, parked, and excluded objects. Parked/excluded objects are shown only to prevent boundary leakage and are not promoted as baseline alternatives.

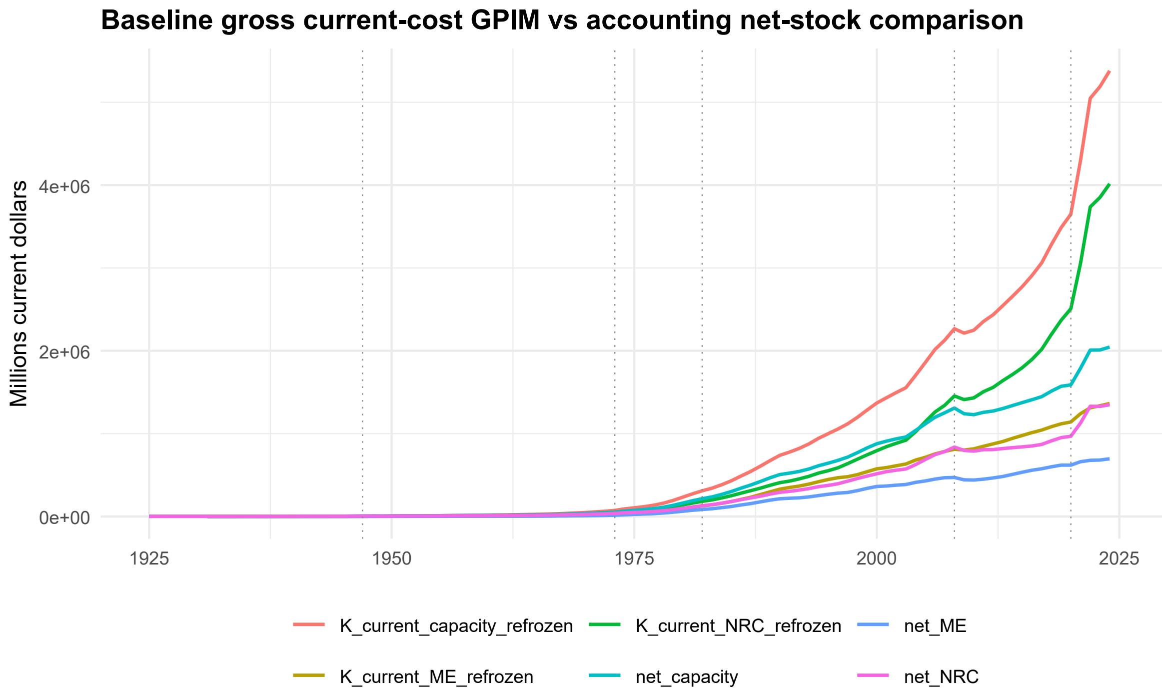

Figure 24: Baseline gross GPIM versus accounting net-stock comparison. Boundary-status diagnostic separating baseline, accounting-comparison, review-only, parked, and excluded objects. Parked/excluded objects are shown only to prevent boundary leakage and are not promoted as baseline alternatives.

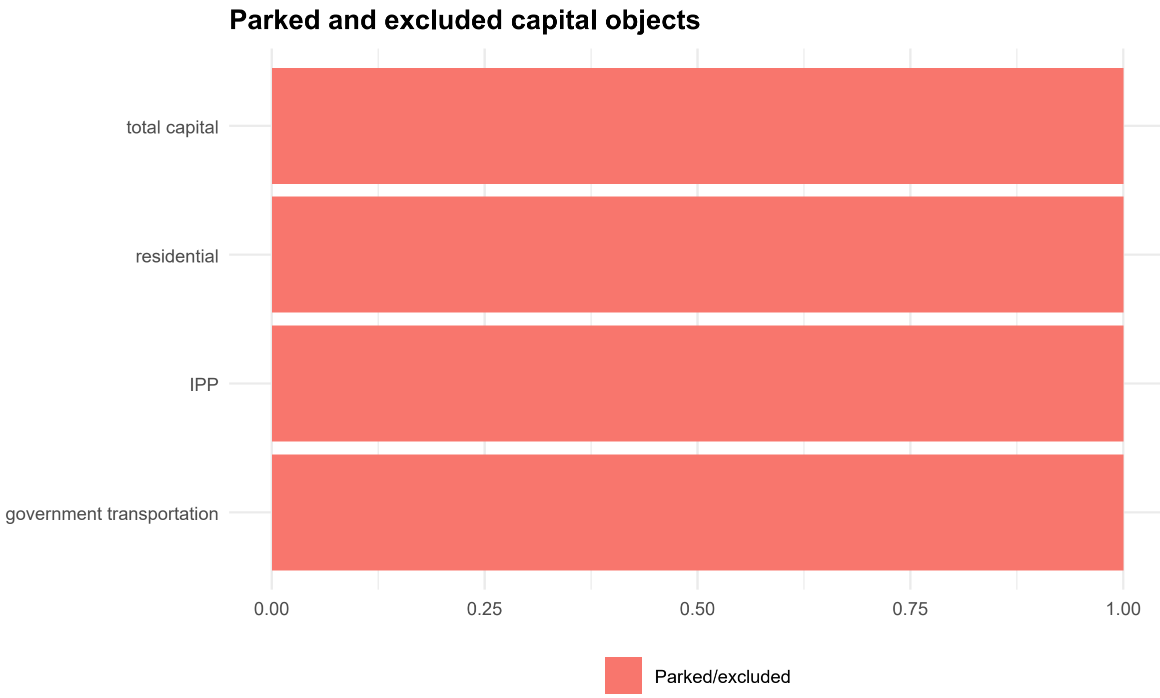

Figure 25: Parked and excluded capital objects status view. Boundary-status diagnostic separating baseline, accounting-comparison, review-only, parked, and excluded objects. Parked/excluded objects are shown only to prevent boundary leakage and are not promoted as baseline alternatives.

11 pKN and Investment-Flow Diagnostics

This section places pKN and investment diagnostics immediately after the capital-status map because pKN is the bridge between real and current-cost GPIM objects. Figures \{}ref{fig:D09_fig26_pKN_ME_NRC_capa \{}ref{fig:D09_fig27_pKN_ME_over_NRC}, \{}ref{fig:D09_fig28_current_vs_real_investment_ME}, \{}ref{fig:D09_fig29_current_vs_real_investment_NRC} show price paths, the ME/NRC price ratio, and current-versus-real guarded investment.

pKN movements help explain why current-cost NRC can behave differently from real NRC. They do not by themselves explain away the real NRC path, which remains a real-stock and service-life review question.

D09-R does not alter pKN, does not recalibrate investment, and does not rebuild the real stock.

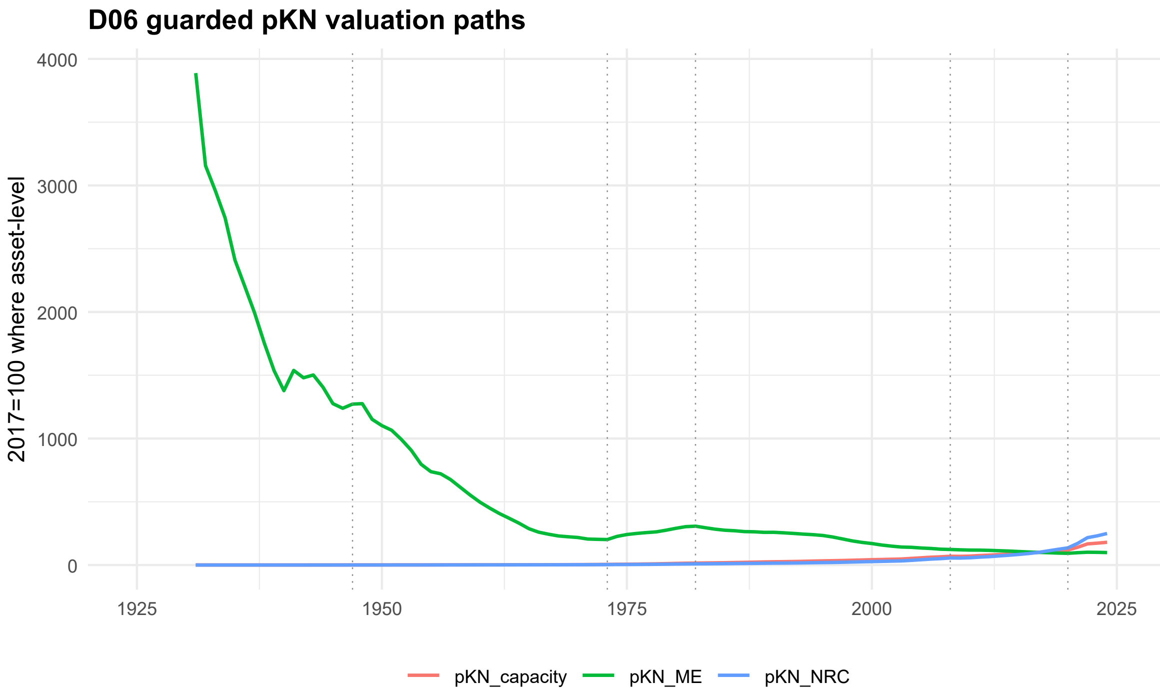

Figure 26: pKN ME, NRC, and capacity paths. D06/D09 pKN and guarded investment-flow diagnostic. The figure helps separate real quantity movements from valuation movements but does not alter pKN or refreeze GPIM stocks.

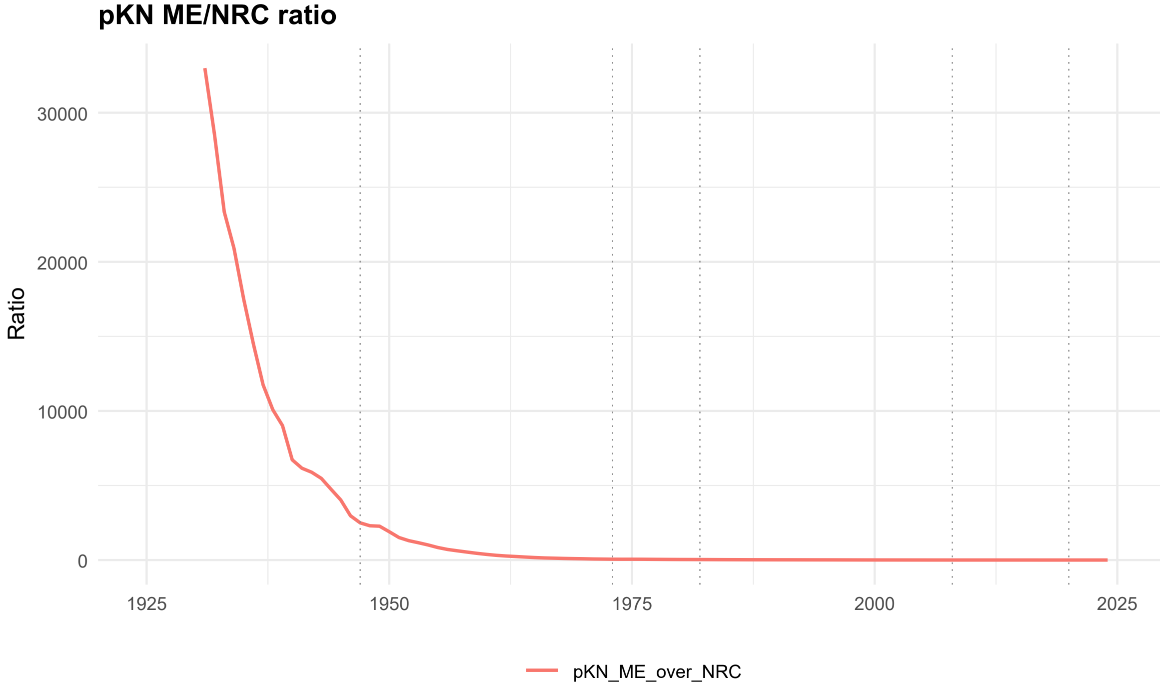

Figure 27: pKN ME/NRC ratio. D06/D09 pKN and guarded investment-flow diagnostic. The figure helps separate real quantity movements from valuation movements but does not alter pKN or refreeze GPIM stocks.

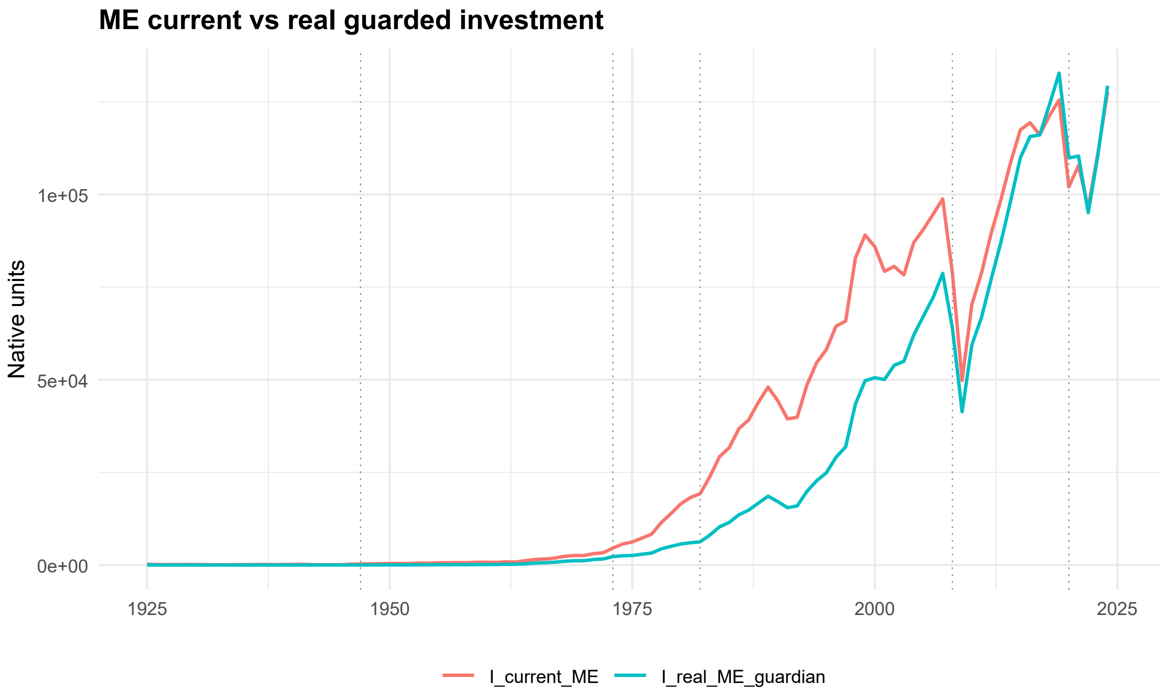

Figure 28: ME current versus guarded real investment. D06/D09 pKN and guarded investment-flow diagnostic. The figure helps separate real quantity movements from valuation movements but does not alter pKN or refreeze GPIM stocks.

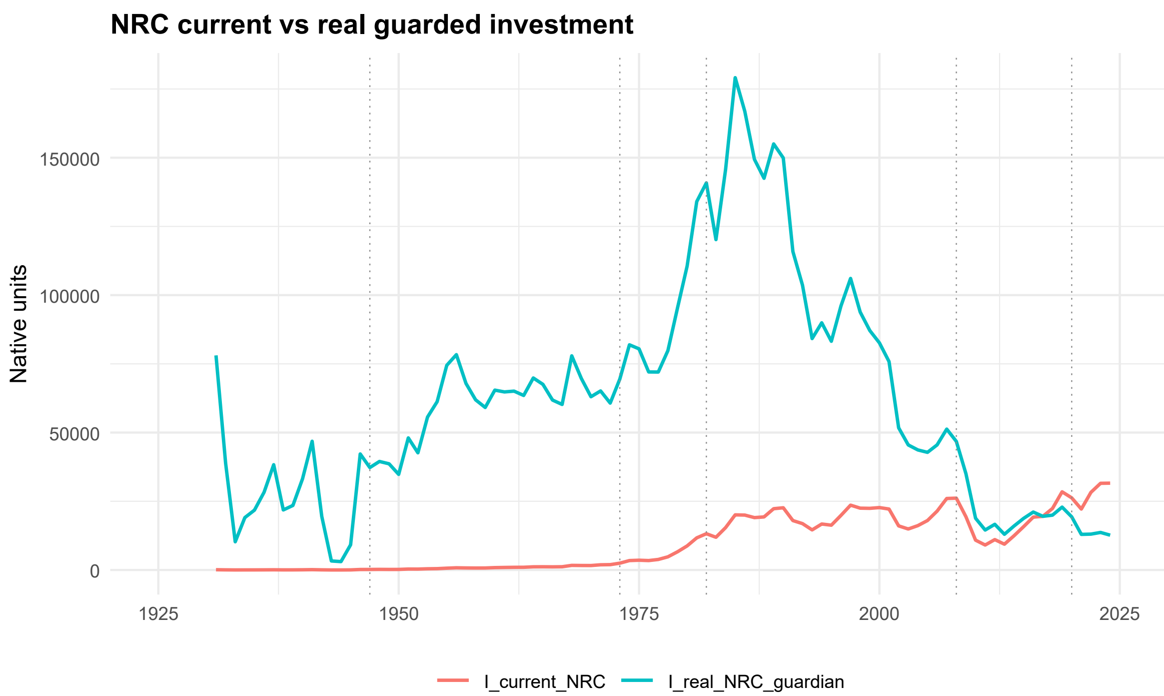

Figure 29: NRC current versus guarded real investment. D06/D09 pKN and guarded investment-flow diagnostic. The figure helps separate real quantity movements from valuation movements but does not alter pKN or refreeze GPIM stocks.

12 Output-Capital Level Relations

This section restores output-capital figures to the main analytical flow before the human checklist. Figures \{}ref{fig:D09_fig30_real_output_and_real_capacity_levels}, \{}ref{fig:D09_fig31_real_output_capacity_
\{}ref{fig:D09_fig32_real_capacity_output_ratio}, \{}ref{fig:D09_fig33_nominal_NFC_GVA_and_current_ca
\{}ref{fig:D09_fig34_nominal_NFC_GVA_current_capacity_ratio}, \{}ref{fig:D09_fig35_scatter_Yreal_NFC_
\{}ref{fig:D09_fig36_scatter_Yreal_NFC_GVA_vs_Kreal_ME}, \{}ref{fig:D09_fig37_scatter_Yreal_NFC_G
show level relations, ratios, and scatterplots for output and capital.

The plots are useful for human plausibility checks: they show whether output and capacity levels move in a way that a later level-analysis design can inspect. There are no fitted lines and no stationarity or cointegration claims.

These figures do not authorize a transformation of μ_t, K, output, or any ratio for model use.

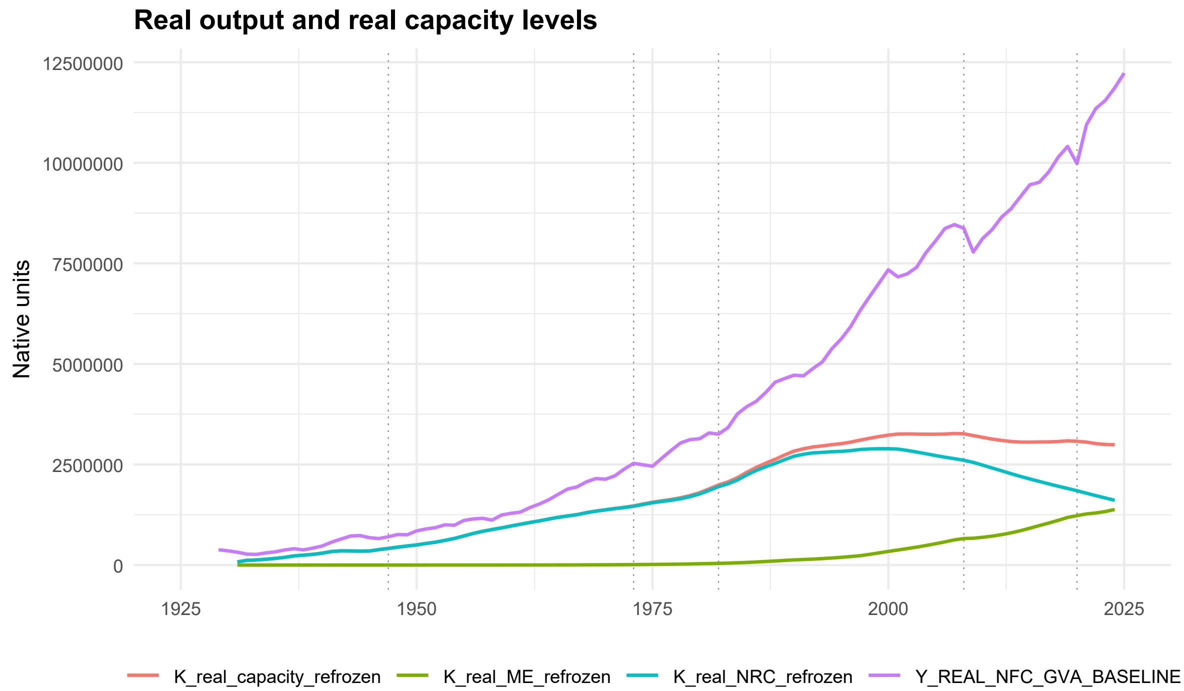

Figure 30: Real output and real capacity levels. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

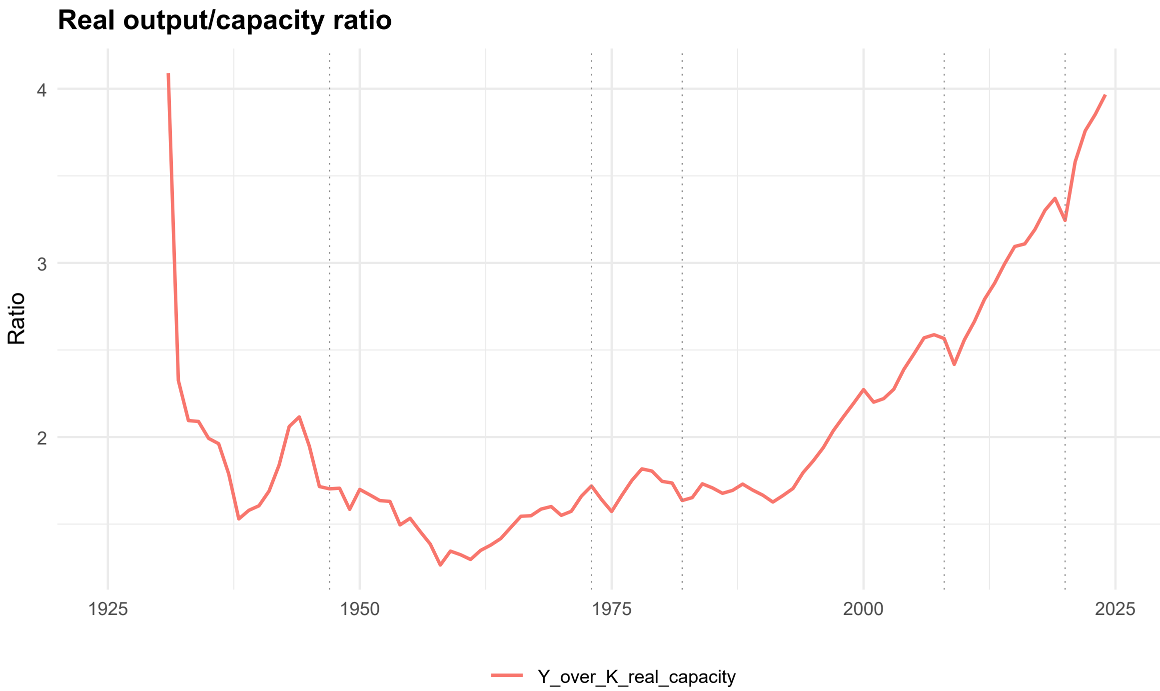

Figure 31: Real output/capacity ratio. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

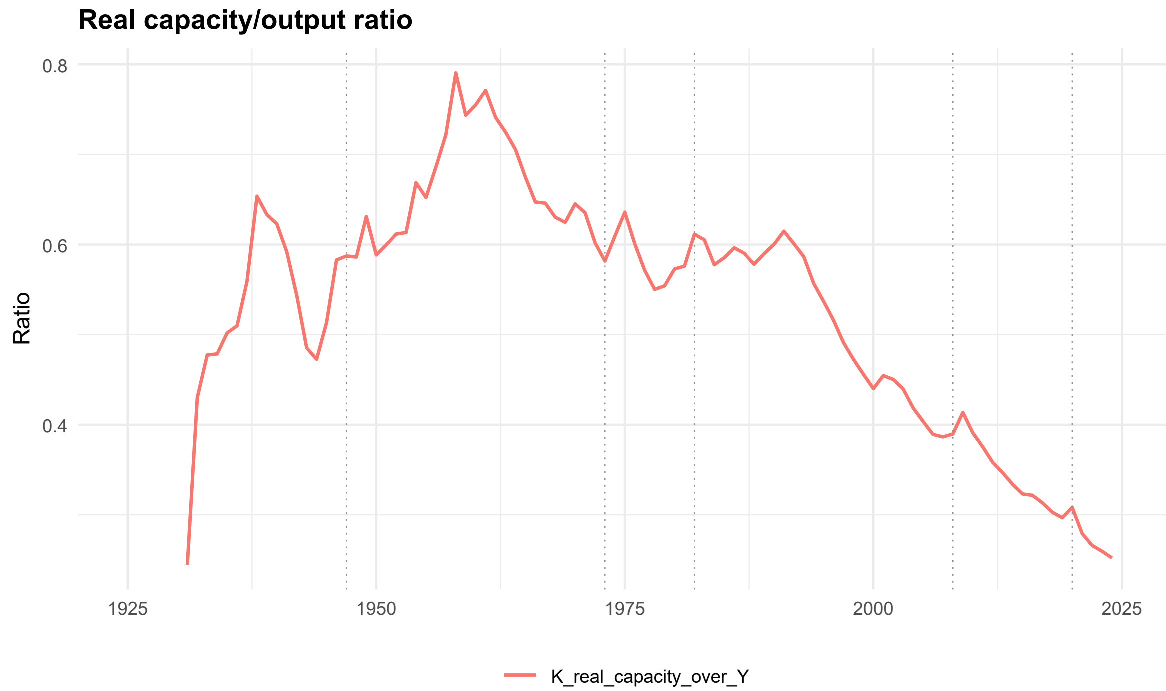

Figure 32: Real capacity/output ratio. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

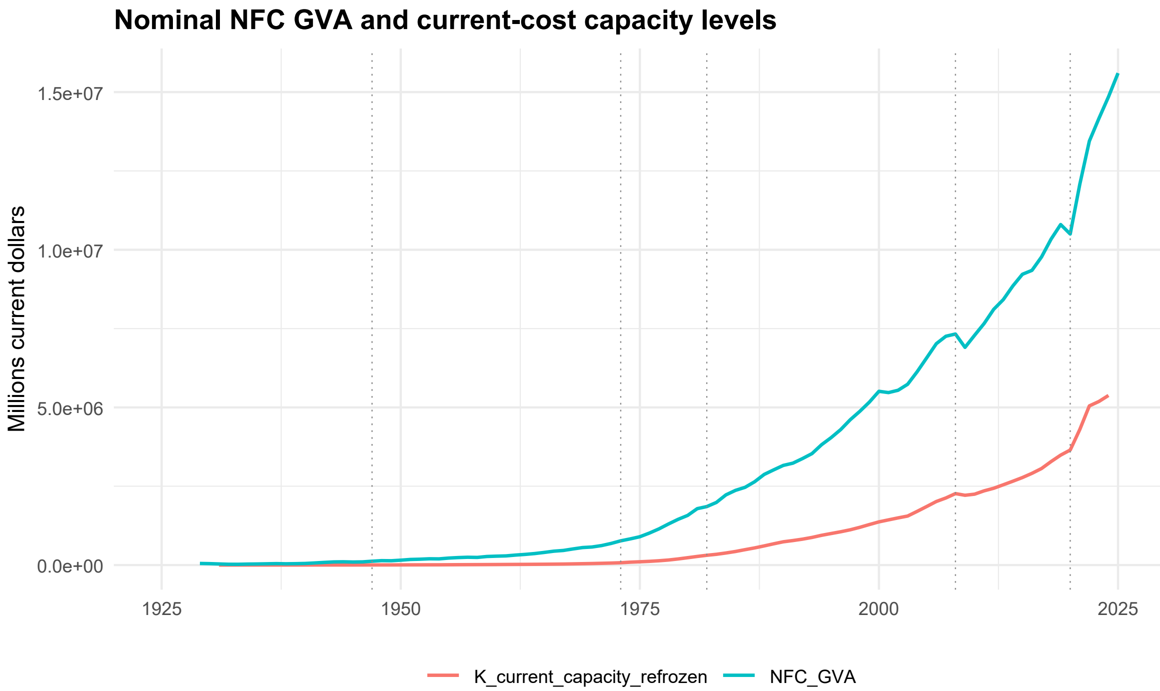

Figure 33: Nominal NFC GVA and current-cost capacity levels. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

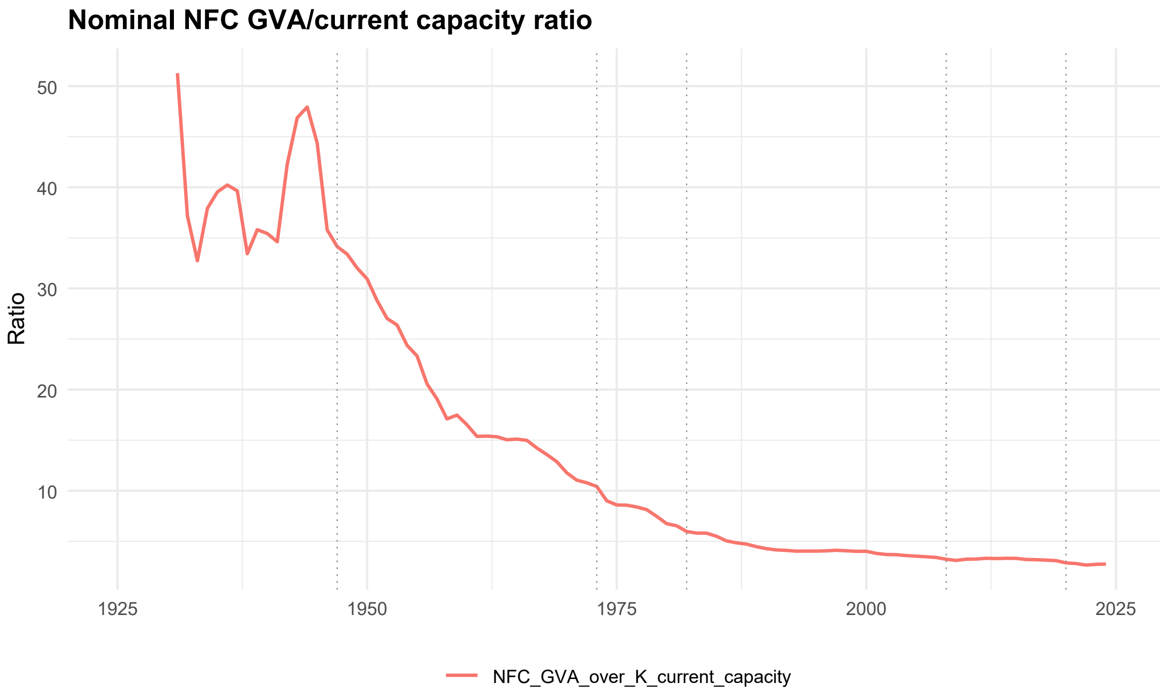

Figure 34: Nominal NFC GVA/current-capacity ratio. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

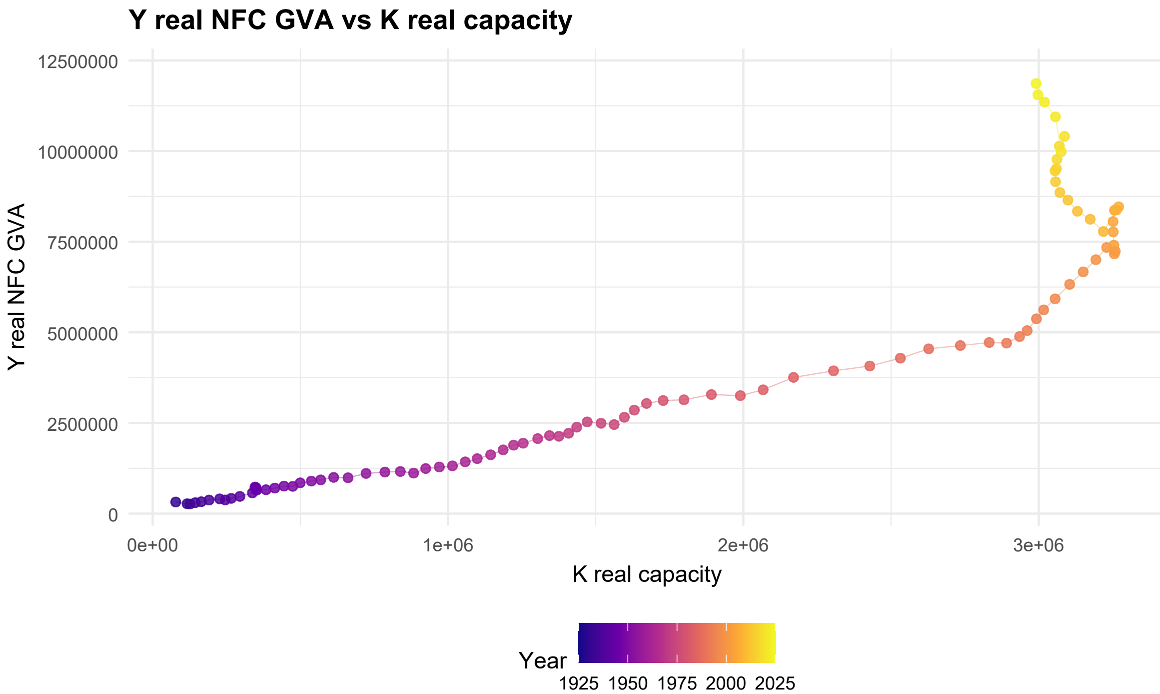

Figure 35: Y real NFC GVA versus K real capacity. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

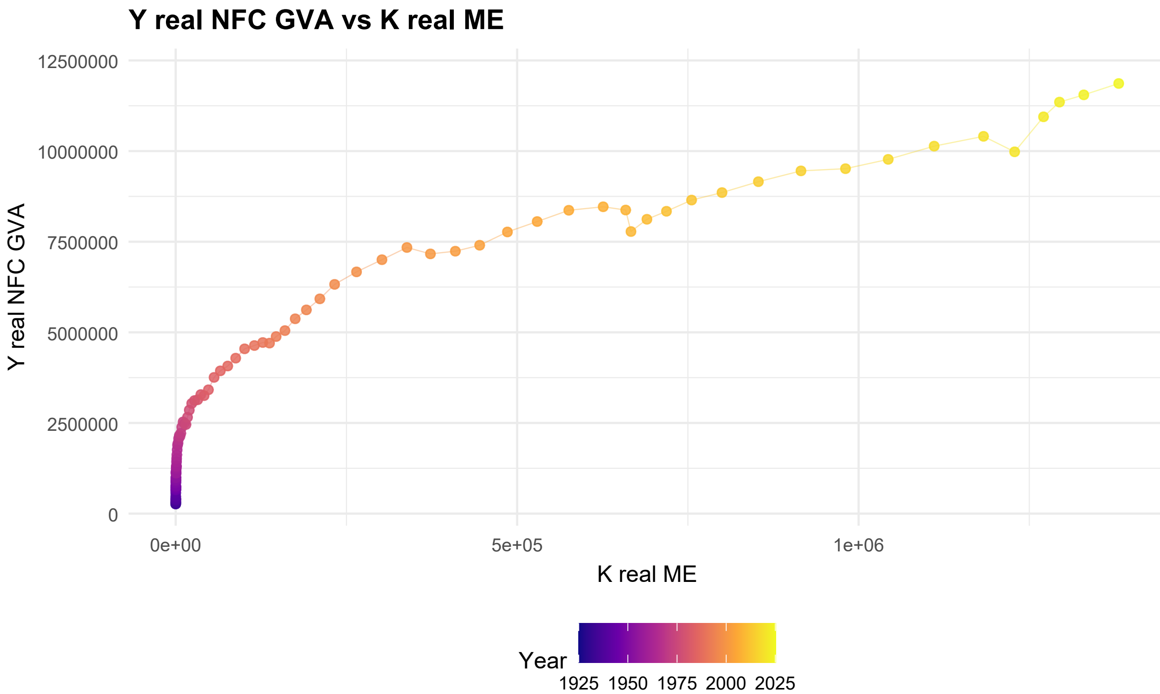

Figure 36: Y real NFC GVA versus K real ME. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

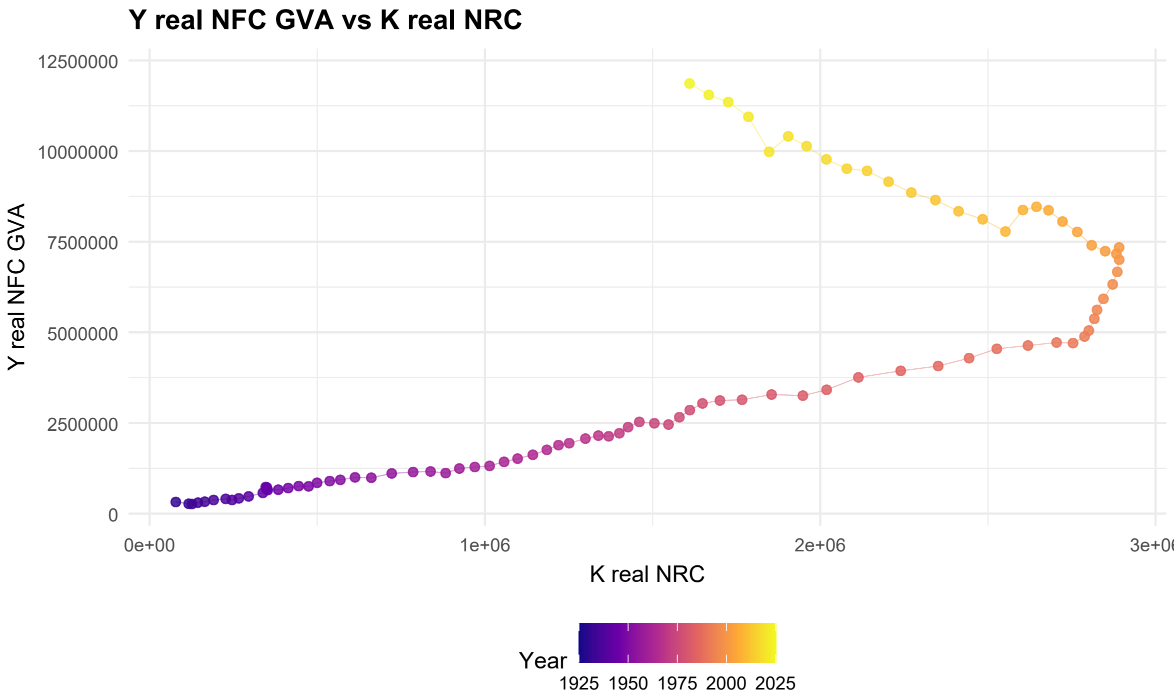

Figure 37: Y real NFC GVA versus K real NRC. D07/D09 level-relation inspection for output and capital. Ratios and scatterplots are REPORT_ONLY, INSPECTION_ONLY, and contain no fitted line, coefficient, stationarity test, or transformation authorization.

13 Distribution and Surplus-Share Inspection

This section keeps all distribution figures in the distribution section rather than orphaning them in an appendix. Figures \{}ref{fig:D09_fig38_NFC_wage_share_variants}, \{}ref{fig:D09_fig39_NFC_gross_net_sur \{}ref{fig:D09_fig40_NFC_CFC_share}, \{}ref{fig:D09_fig41_CORP_wage_share_reconciliation_variants}, \{}ref{fig:D09_fig42_CORP_surplus_share_reconciliation_variants} show wage, surplus, CFC, and corporate reconciliation variants.

The figures preserve the D07 status distinctions between baseline shares, reconciliation variants, and report-only scaffolds. This keeps surplus-share inspection inside the accounting boundary.

No distribution series is promoted beyond its D07 authorization status, and no transformation use is authorized.

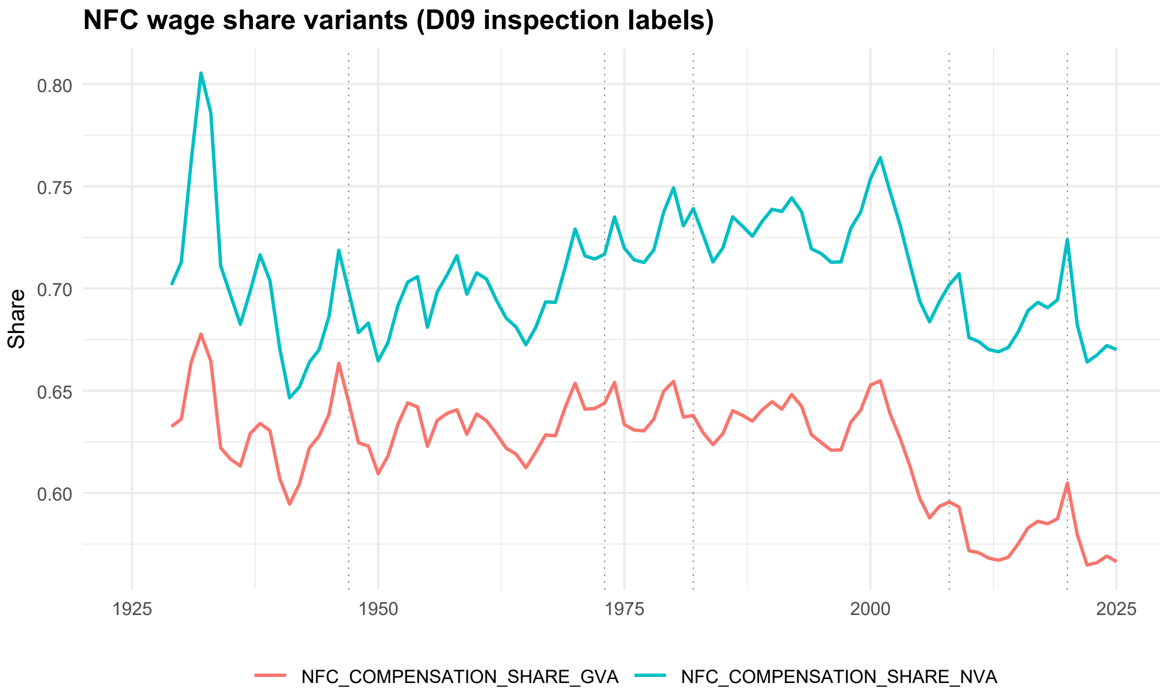

Figure 38: NFC wage share variants. D07/D09 distribution inspection figure. Authorized baseline shares remain distinguished from reconciliation and report-only scaffolds; no distribution variable is promoted beyond its D07 status.

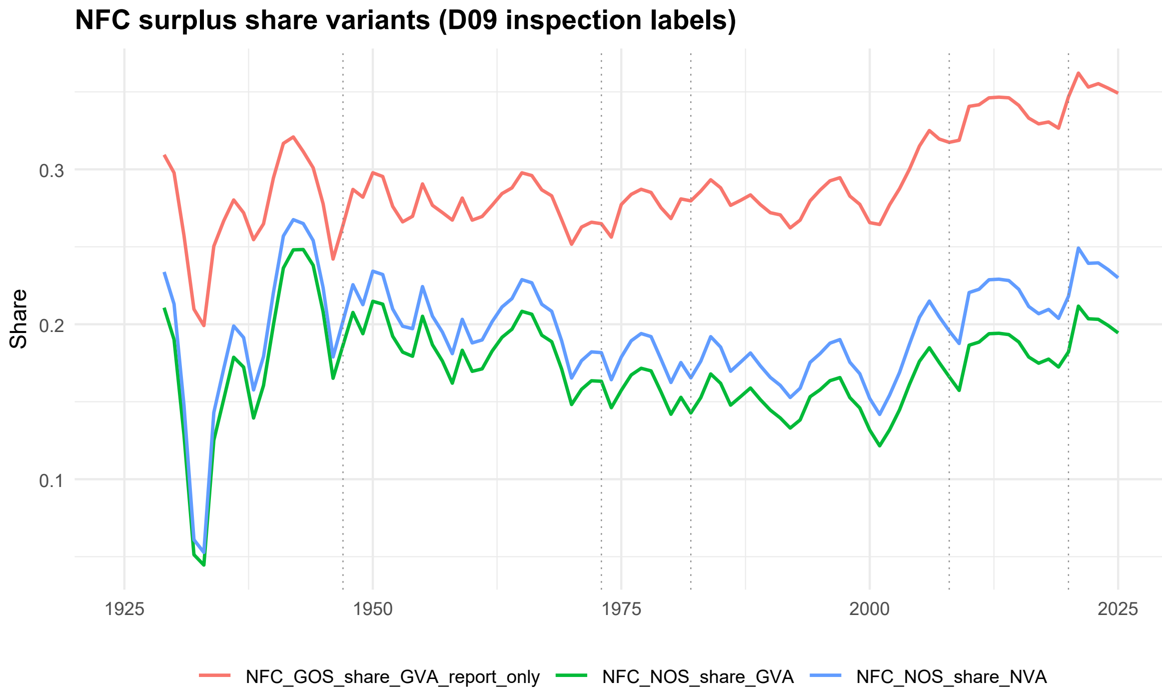

Figure 39: NFC surplus share variants. D07/D09 distribution inspection figure. Authorized baseline shares remain distinguished from reconciliation and report-only scaffolds; no distribution variable is promoted beyond its D07 status.

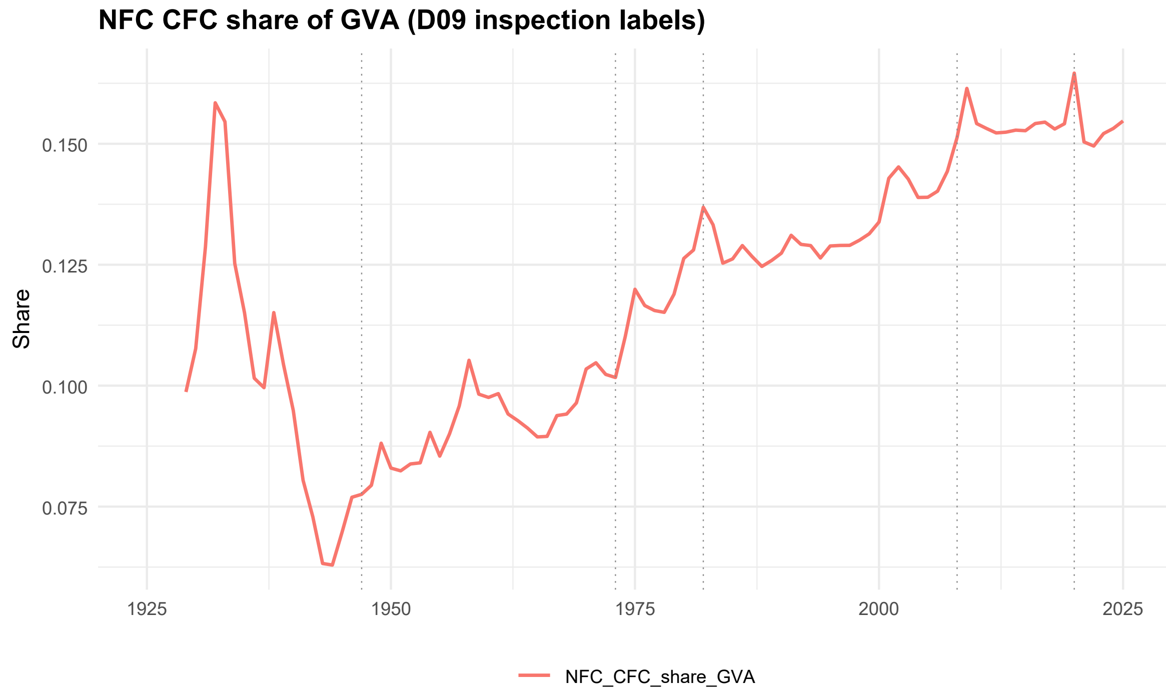

Figure 40: NFC CFC share. D07/D09 distribution inspection figure. Authorized baseline shares remain distinguished from reconciliation and report-only scaffolds; no distribution variable is promoted beyond its D07 status.

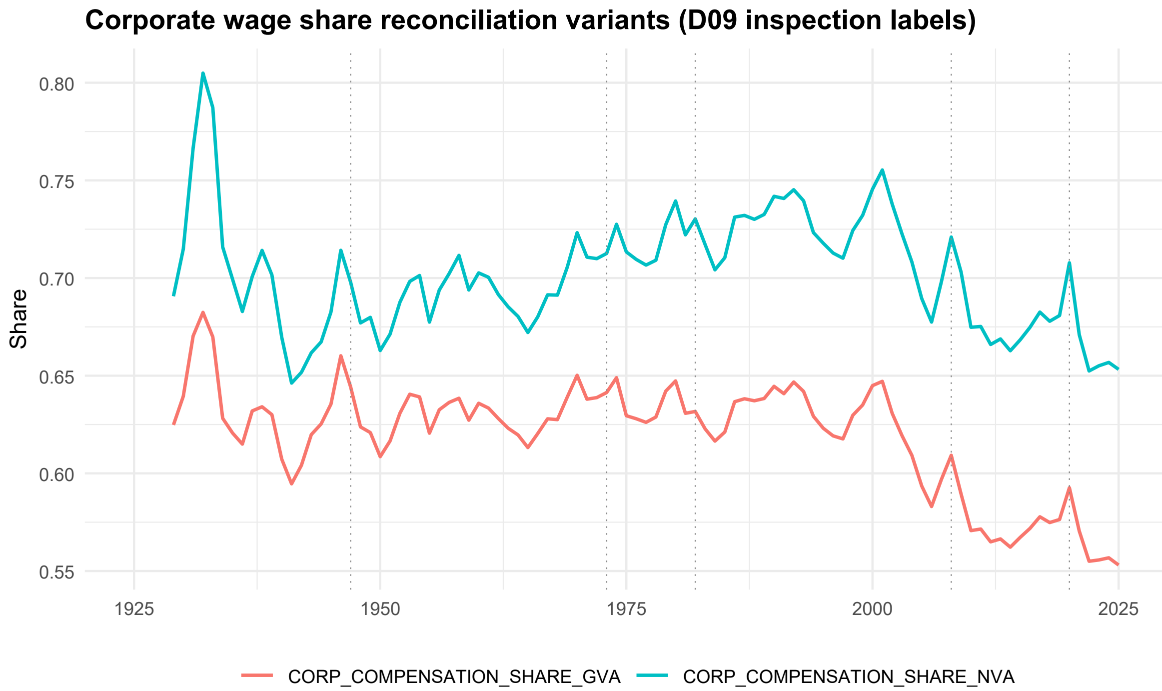

Figure 41: Corporate wage share reconciliation variants. D07/D09 distribution inspection figure. Authorized baseline shares remain distinguished from reconciliation and report-only scaffolds; no distribution variable is promoted beyond its D07 status.

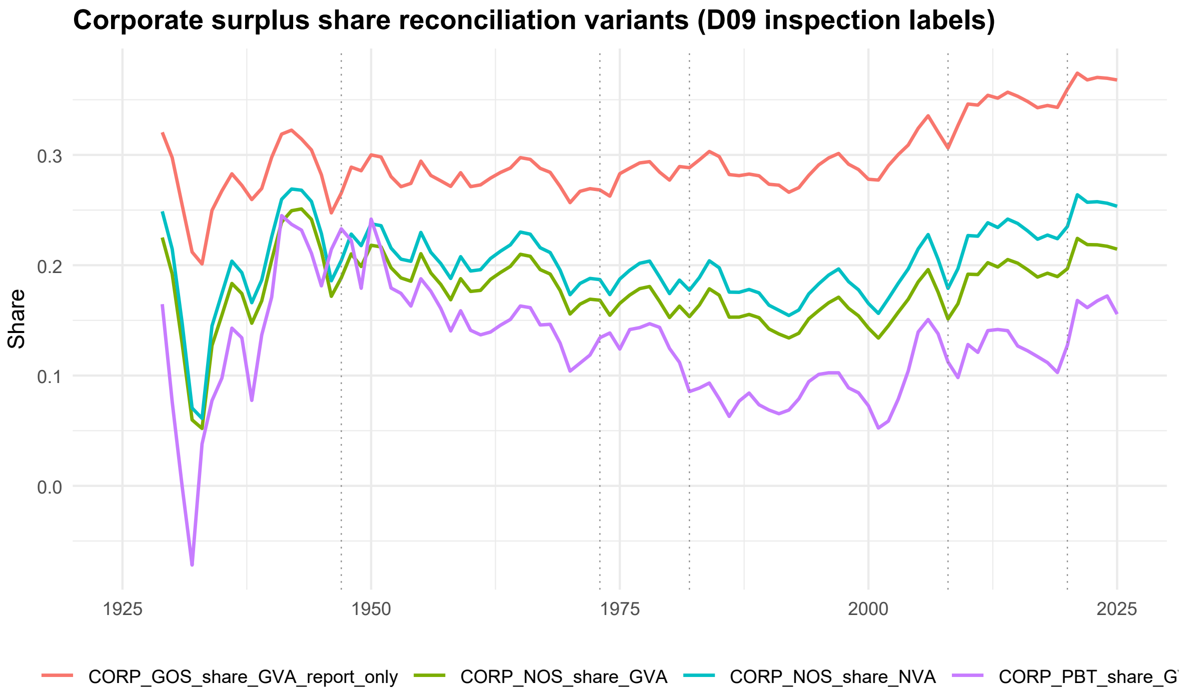

Figure 42: Corporate surplus share reconciliation variants. D07/D09 distribution inspection figure. Authorized baseline shares remain distinguished from reconciliation and report-only scaffolds; no distribution variable is promoted beyond its D07 status.

14 Financial-Sector Transfer and Double-Counting Safeguards

This section places the financial-transfer safeguard where it is interpreted. Figure \{}ref{fig:D09_fig43_surplus_acco is read after the table-first safeguard statement.

The financial ladder is a boundary device: productive-origin surplus, accounting reconciliation, financial-transfer layers, and imputed-interest corrections must not be silently collapsed into one profit object.

The section does not authorize financial correction as baseline productive-origin surplus.

layer	status	safeguard
NFC productive-origin baseline	authorized baseline/accounting	productive-origin baseline
Corporate reconciliation variants	authorized reconciliation	comparison only
Financial transfer/correction candidates	candidate only	not silently productive-origin surpl
Imputed-interest candidates	candidate only	requires semantic crosswalk

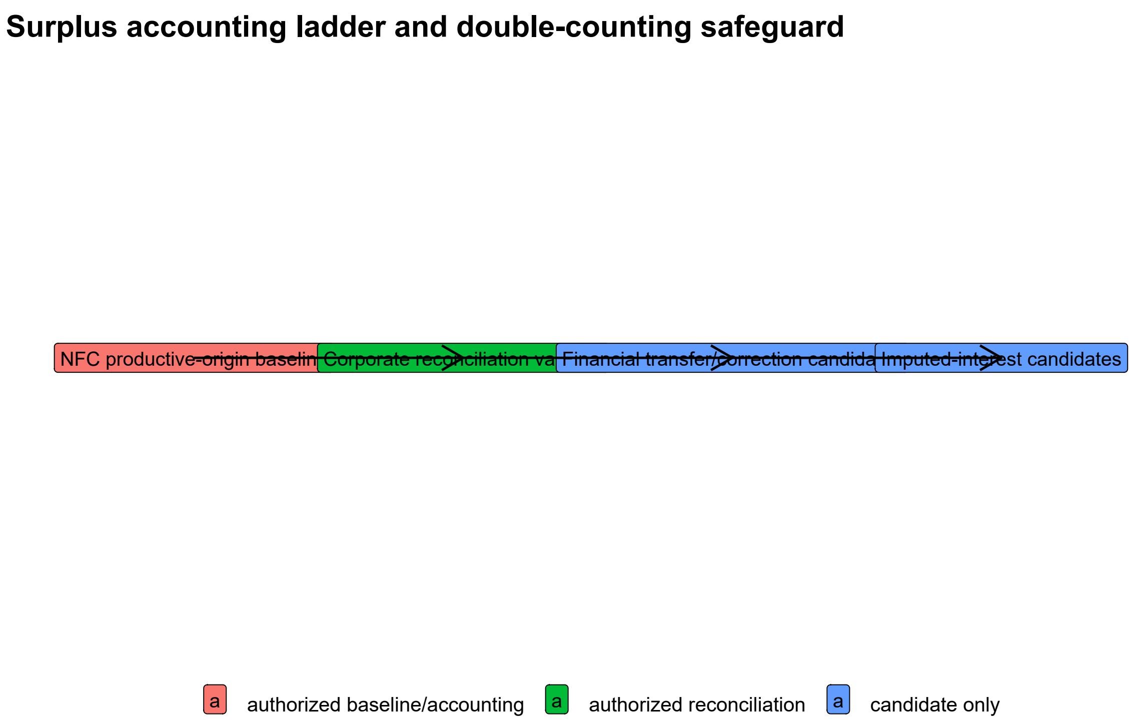

Figure 43: Surplus accounting ladder and double-counting safeguard. Financial-transfer safeguard diagnostic from D09. The table-first interpretation separates productive-origin surplus from financial correction and imputed-interest layers, preventing double counting or boundary leakage.

15 Capital-Output-Distribution Relational Inspection

This section collects the relational scatterplots after their ingredients have been shown. Figures \{}ref{fig:D09_fig44_wage_share_vs_ME_over_NRC}, \{}ref{fig:D09_fig45_wage_share_vs_ME_share_of_c \{}ref{fig:D09_fig46_wage_share_vs_capacity_output_ratio}, \{}ref{fig:D09_fig47_surplus_share_vs_ME_ov \{}ref{fig:D09_fig48_surplus_share_vs_capacity_output_ratio} relate distribution shares to ME/NRC and capacity-output diagnostics.

These plots are visual inspection only. The absence of fitted lines is intentional: D09-R is not estimating a relation or testing a model.

No coefficient, fitted relation, or transformation is authorized here.

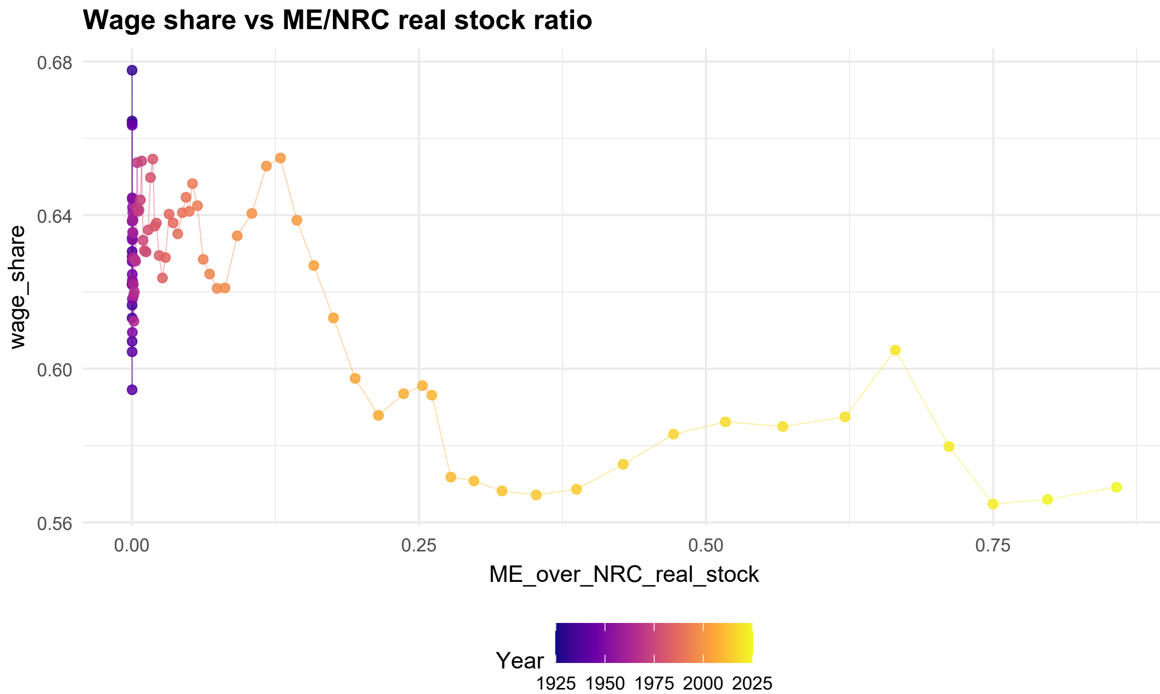

Figure 44: Wage share versus ME/NRC real stock ratio. D09 report-only relational inspection. The figure is visual only, uses no fitted line, and does not authorize transformation or econometric use.

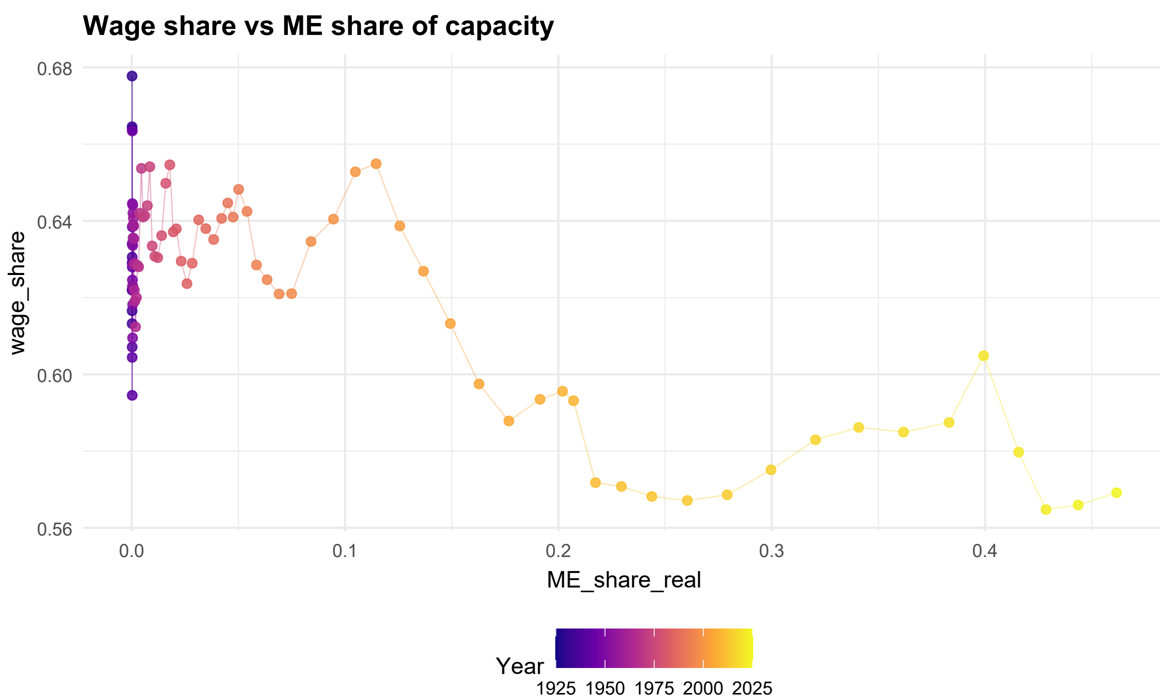

Figure 45: Wage share versus ME share of capacity. D09 report-only relational inspection. The figure is visual only, uses no fitted line, and does not authorize transformation or econometric use.

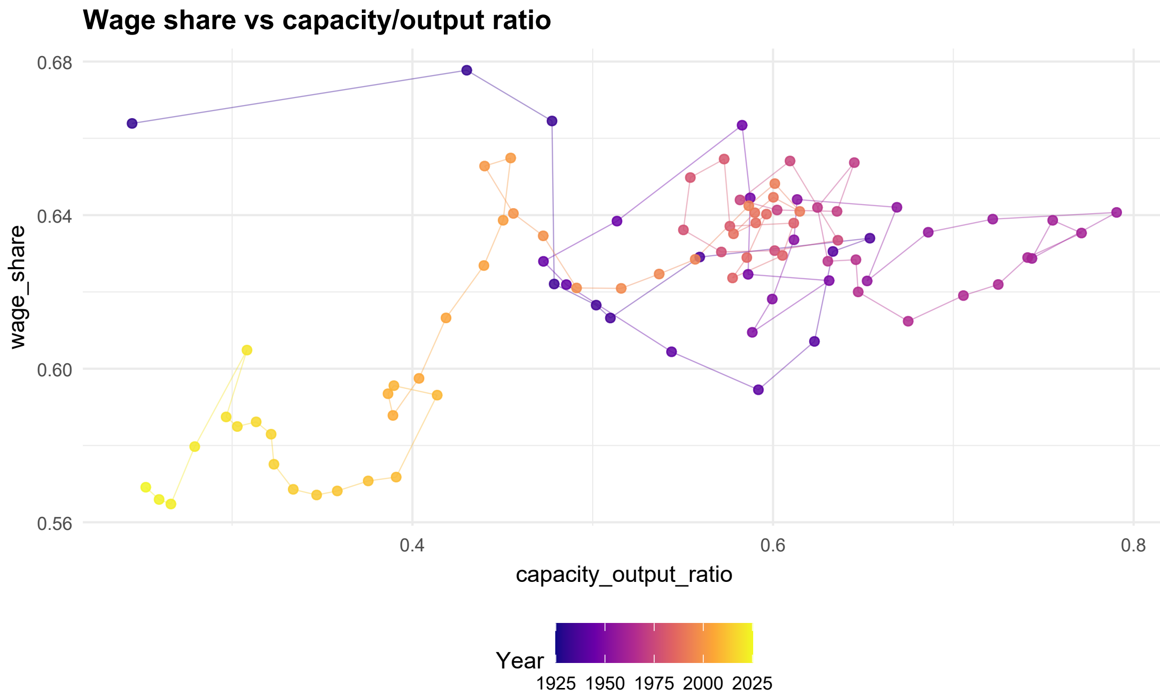

Figure 46: Wage share versus capacity/output ratio. D09 report-only relational inspection. The figure is visual only, uses no fitted line, and does not authorize transformation or econometric use.

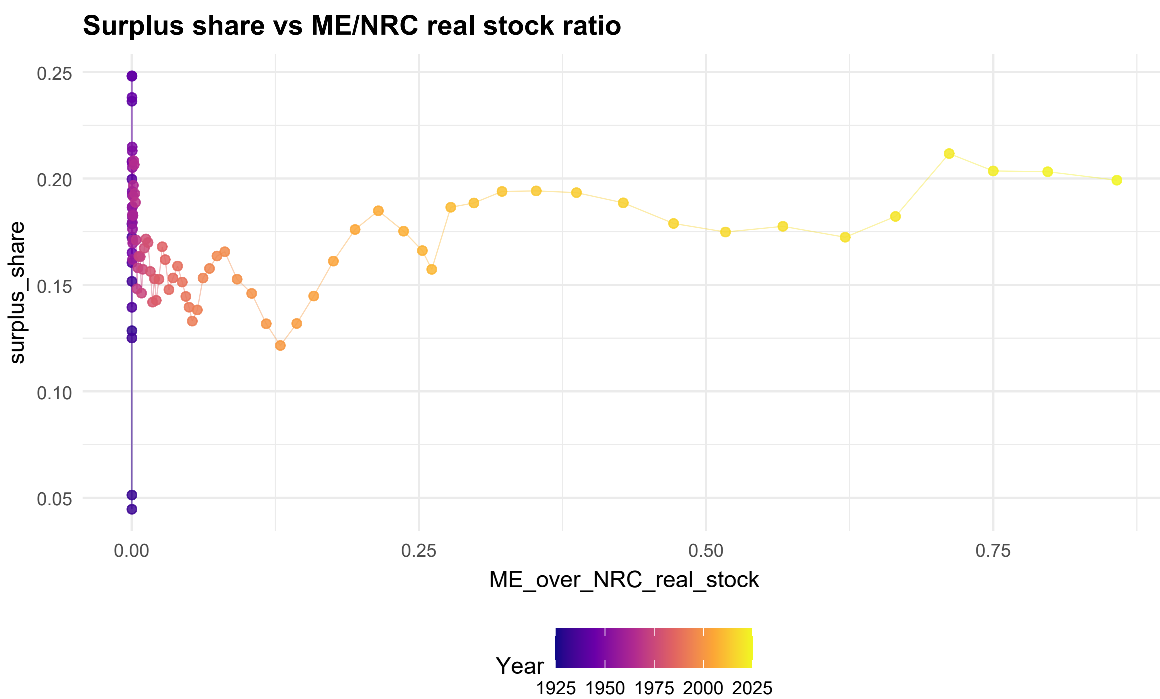

Figure 47: Surplus share versus ME/NRC real stock ratio. D09 report-only relational inspection. The figure is visual only, uses no fitted line, and does not authorize transformation or econometric use.

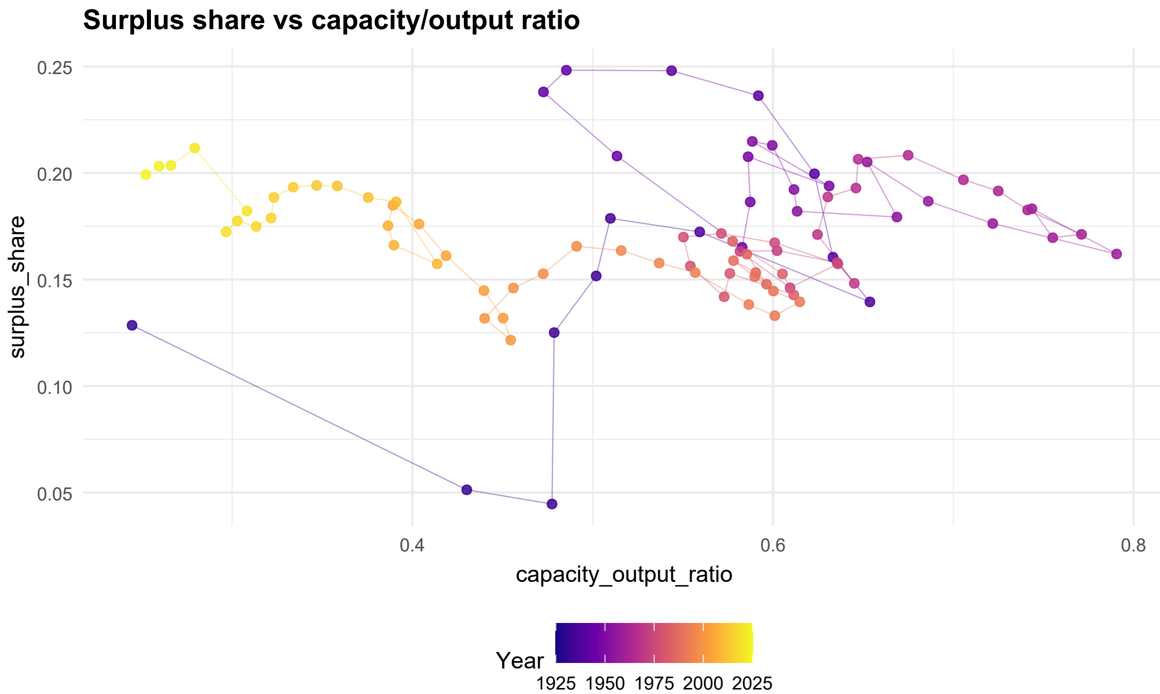

Figure 48: Surplus share versus capacity/output ratio. D09 report-only relational inspection. The figure is visual only, uses no fitted line, and does not authorize transformation or econometric use.

16 Human Interpretation of the NRC Decline

The NRC decline can be defended only with the correct object definition. The object is nonfinancial productive-capacity nonresidential construction, not total nonresidential buildings and not all standing structures.

The defensible interpretation is boundary movement: abandonment, rezoning, conversion to residential use, or absorption into financial/real-estate uses can remove structures from the productive-capacity object without literal demolition. That interpretation is plausible, but the L=30 service-life assumption and short NRC warmup leave a medium review-required risk.

This section does not resolve the service-life issue. It makes the issue explicit and prevents the report from overstating the physical meaning of the real NRC stock.

17 Future Sensitivity Design: NRC Service Life and Warmup

This section records a future sensitivity design without executing it. The proposed pass would compare longer NRC service lives and possibly a longer infrastructure-style survival profile if literature supports it.

The future design keeps the baseline intact: NRC L=30, alpha=1.6 remains the D06/D07/D09 object. Alternatives such as L=40, L=50, and L=60 are proposed future report-only cases, not D09-R results.

D09-R does not run sensitivity analysis and does not authorize D10 transformation planning.

reference_key	author_year	asset_s
Nomura2005_NRC_service_life_REVIEW	Nomura 2005	nonresid
BEA_fixed_assets_service_life_documentation_REVIEW	BEA fixed assets documentation	fixed as
OECD_capital_measurement_manual_REVIEW	OECD capital measurement manual	capital
Hulten_Wykoff_depreciation_literature_REVIEW	Hulten-Wykoff depreciation literature	deprecia
local_repo_GPIM_service_life_context_REVIEW	Local repo context	D01-D0

18 Human Decision Checklist

The checklist records the human-facing outcome after layout repair and NRC interpretation repair.

The report is ready for D09 double validation review with the NRC service-life issue carried as medium review-required.

The checklist does not authorize model transformations or D10 implementation.

item	status
Revised report figure placement is aligned with section text	PASS
NRC decline interpretation is explicit and boundary-safe	PASS
NRC L=30 and warmup concern are carried forward	REVIEW_REQUIRED
No GPIM rebuild, survival change, pKN change, or econometrics are run	PASS
Future sensitivity is proposed but not executed	PASS
Ready for D09 double validation review	PASS_WITH_REVIEW_FLAG

A Secondary Figures

No D09 figure is orphaned in the appendix in this repaired report. The parked/excluded status view appears in Section 10 because it is needed to explain boundary status, not because it is a baseline alternative.

B Audit Tables

The revised validation checks, human review flags, NRC interpretation ledger, and service-life literature placeholders are written to the D09-R csv and tables folders. Large ledgers remain CSV-first to avoid overfull main-body tables.

C Figure Manifest

The revised figure-placement ledger records one placement decision for each of the 48 copied D09 figures. The source figure manifest remains D09_figure_manifest.csv; the repaired placement ledger is D09_R_figure_placement_ledger.csv.